

Lemon Bay Aquatic Preserve

SEACAR Habitat Analyses

Last compiled on 10 June, 2026

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Funding & Acknowledgements

The data used in this analysis is from the Export Standardized Tables in the SEACAR Data Discovery Interface (DDI). Documents and information available through the SEACAR DDI are owned by the data provider(s) and users are expected to provide appropriate credit following accepted citation formats. Users are encouraged to access data to maximize utilization of gained knowledge, reducing redundant research and facilitating partnerships and scientific innovation.

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Threshold Filtering

Threshold filters, following the guidance of Florida Department of Environmental Protection's (*FDEP*) Division of Environmental Assessment and Restoration (*DEAR*) are used to exclude specific results values from the SEACAR Analysis. Based on the threshold filters, Quality Assurance / Quality Control (*QAQC*) Flags are inserted into the *SEACAR_QAQCFlagCode* and *SEACAR_QAQC_Description* columns of the export data. The *Include* column indicates whether the *QAQC* Flag will also indicate that data are excluded from analysis. No data are excluded from the data export, but the analysis scripts can use the *Include* column to exclude data (1 to include, 0 to exclude).

Table 1: Continuous Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Dissolved Oxygen	mg/L	-0.000001	50
Dissolved Oxygen Saturation	%	-0.000001	500
Fluorescent Dissolved Organic Matter	QSE	-	-
Salinity	ppt	-0.000001	70
Specific Conductivity	mS/cm	-0.000001	200
Turbidity	NTU	-0.000001	4000
Water Temperature	Degrees C	-5.000000	45
pH	None	2.000000	14

Table 2: Discrete Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Ammonia, Un-ionized (NH3)	mg/L	-	-
Ammonium (NH4)	mg/L	-	-

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Corrected for Pheophytin	ug/L	-	-
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Colored Dissolved Organic Matter	PCU	-	-
Dissolved Oxygen	mg/L	-0.000001	25
Dissolved Oxygen Saturation	%	-0.000001	310
Fluorescent Dissolved Organic Matter	QSE	-	-
Light Extinction Coefficient	m ⁻¹	-	-
NO2+3, Filtered	mg/L	-	-
Nitrate (NO3)	mg/L	-	-
Nitrite (NO2)	mg/L	-	-
Nitrogen, inorganic	mg/L	-	-
Nitrogen, organic	mg/L	-	-
Phosphate, Filtered (PO4)	mg/L	-	-
Salinity	ppt	-0.000001	70
Secchi Depth	m	0.000001	50
Specific Conductivity	mS/cm	0.005000	100
Total Ammonia (N)	mg/L	-	-
Total Kjeldahl Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Phosphorus	mg/L	-	-
Total Suspended Solids	mg/L	-	-
Turbidity	NTU	-	-
Water Temperature	Degrees C	3.000000	40
pH	None	2.000000	13

Table 3: Quality Assurance Flags inserted based on threshold checks listed in Table 1 and 2

<i>SEACAR QAQC Description</i>	<i>Include</i>	<i>SEACAR QAQCFlagCode</i>
Exceeds maximum threshold	0	2Q
Below minimum threshold	0	4Q
Within threshold tolerance	1	6Q
No defined thresholds for this parameter	1	7Q

Value Qualifiers

Value qualifier codes included within the data are used to exclude certain results from the analysis. The data are retained in the data export files, but the analysis uses the *Include* column to filter the results.

STORET and WIN value qualifier codes

Value qualifier codes from *STORET* and *WIN* data are examined with the database and used to populate the *Include* column in data exports.

Table 4: Value Qualifier codes excluded from analysis

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>MDL</i>	<i>Description</i>
STORET-WIN	H	0	0	Value based on field kit determination; results may not be accurate
STORET-WIN	J	0	0	Estimated value
STORET-WIN	V	0	0	Analyte was detected at or above method detection limit
STORET-WIN	Y	0	0	Lab analysis from an improperly preserved sample; data may be inaccurate

Discrete Water Quality Value Qualifiers

The following value qualifiers are highlighted in the Discrete Water Quality section of this report. An exception is made for **Program 476** - *Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network* and data flagged with Value Qualifier **H** are included for this program only.

H - Value based on field kit determination; results may not be accurate. This code shall be used if a field screening test (e.g., field gas chromatograph data, immunoassay, or vendor-supplied field kit) was used to generate the value and the field kit or method has not been recognized by the Department as equivalent to laboratory methods.

I - The reported value is greater than or equal to the laboratory method detection limit but less than the laboratory practical quantitation limit.

Q - Sample held beyond the accepted holding time. This code shall be used if the value is derived from a sample that was prepared or analyzed after the approved holding time restrictions for sample preparation or analysis.

S - Secchi disk visible to bottom of waterbody. The value reported is the depth of the waterbody at the location of the Secchi disk measurement.

U - Indicates that the compound was analyzed for but not detected. This symbol shall be used to indicate that the specified component was not detected. The value associated with the qualifier shall be the laboratory method detection limit. Unless requested by the client, less than the method detection limit values shall not be reported

Systemwide Monitoring Program (SWMP) value qualifier codes

Value qualifier codes from the *SWMP* continuous program are examined with the database and used to populate the *Include* column in data exports. *SWMP* Qualifier Codes are indicated by *QualifierSource=SWMP*.

Table 5: SWMP Value Qualifier codes

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>Description</i>
SWMP	-1	1	Optional parameter not collected
SWMP	-2	0	Missing data
SWMP	-3	0	Data rejected due to QA/QC
SWMP	-4	0	Outside low sensor range
SWMP	-5	0	Outside high sensor range
SWMP	0	1	Passed initial QA/QC checks
SWMP	1	0	Suspect data
SWMP	2	1	Reserved for future use
SWMP	3	1	Calculated data: non-vented depth/level sensor correction for changes in barometric pressure
SWMP	4	1	Historical: Pre-auto QA/QC
SWMP	5	1	Corrected data

Water Column

The water column habitat extends from the water's surface to the bottom sediments, and it's where fish, dolphins, crabs and people swim! So much life makes its home in the water column that the health of marine and coastal ecosystems, as well as human economies, depend on the condition of this vulnerable habitat. Local patterns of rainfall, temperature, winds and currents can rapidly change the condition of the water column, while global influences such as [El Niño/La Niña](#), large-scale fluctuation in sea temperatures and climate change can have long-term effects. Inputs from the prosperity of our day-to-day lives including farming, mining and forestry, and emissions from power generation, automobiles and water treatment can also alter the health of the water column. Acting alone or together, each input can have complex and lasting effects on habitats and ecosystems.

SEACAR evaluates water column health with several essential parameters. These include nutrient surveys of nitrogen and phosphorus, and water quality assessments of salinity, dissolved oxygen, pH, and water temperature. Water clarity is evaluated with Secchi depth, turbidity, levels of chlorophyll a, total suspended solids, and colored dissolved organic matter. Additionally, the richness of nekton is indicated by the abundance of free-swimming fishes and macroinvertebrates like crabs and shrimps.

Seasonal Kendall-Tau Analysis

Indicators must have a minimum of five to ten years, depending on the habitat, of data within the geographic range of the analysis to be included in the analysis. Ten years of data are required for discrete parameters, and five years of data are required for continuous parameters. If there are insufficient years of data, the number of years of data available will be noted and labeled as "insufficient data to conduct analysis". Further, for the preferred Seasonal Kendall-Tau test, there must be data from at least two months in common across at least two consecutive years within the RCP managed area being analyzed. Values that pass both of these tests will be included in the analysis and be labeled as *Use_In_Analysis* = **TRUE**. Any that fail either test will be excluded from the analyses and labeled as *Use_In_Analysis* = **FALSE**. The points for all Water Column plots displayed in this section are monthly averages. Trend significance will be denoted as "Significant Trend" (when $p < 0.05$), or "Non-significant Trend" (when $p \geq 0.05$). Any parameters with insufficient data to perform Seasonal Kendall-Tau test will have their monthly averages plotted without a corresponding trend line.

Water Quality - Discrete

The following files were used in the discrete analysis:

- *Combined_WQ_WC_NUT_Chlorophyll_a_corrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Chlorophyll_a_uncorrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Colored_dissolved_organic_matter_CDOM-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen_Saturation-2026-May-18.txt*
- *Combined_WQ_WC_NUT_pH-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Salinity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Secchi_Depth-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Nitrogen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Phosphorus-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Suspended_Solids_TSS-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Turbidity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Water_Temperature-2026-May-18.txt*

Chlorophyll a, Corrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

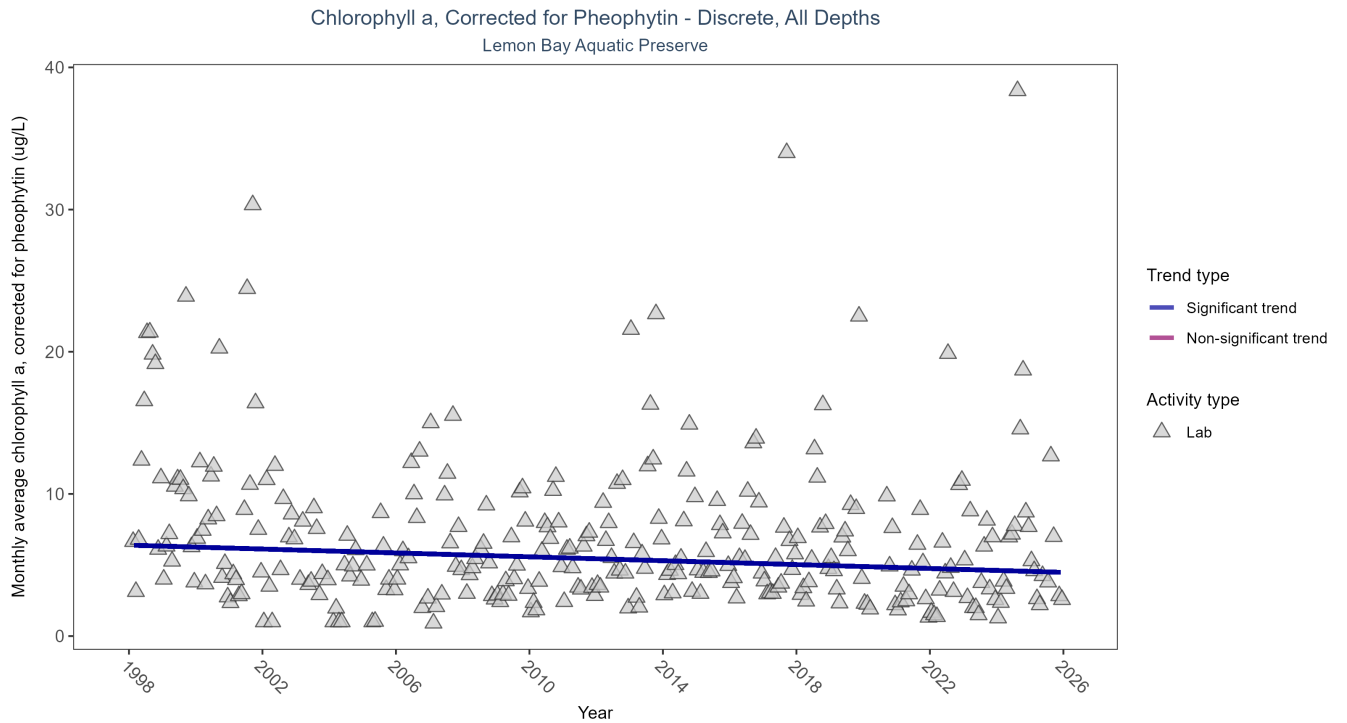


Figure 1: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 6: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	1278	28	1998 - 2025	4.865	-0.1373	6.3968	-0.0684	0.0008

Monthly average chlorophyll a, corrected for pheophytin, decreased by 0.07 $\mu\text{g/L}$ per year, indicating an increase in water clarity.

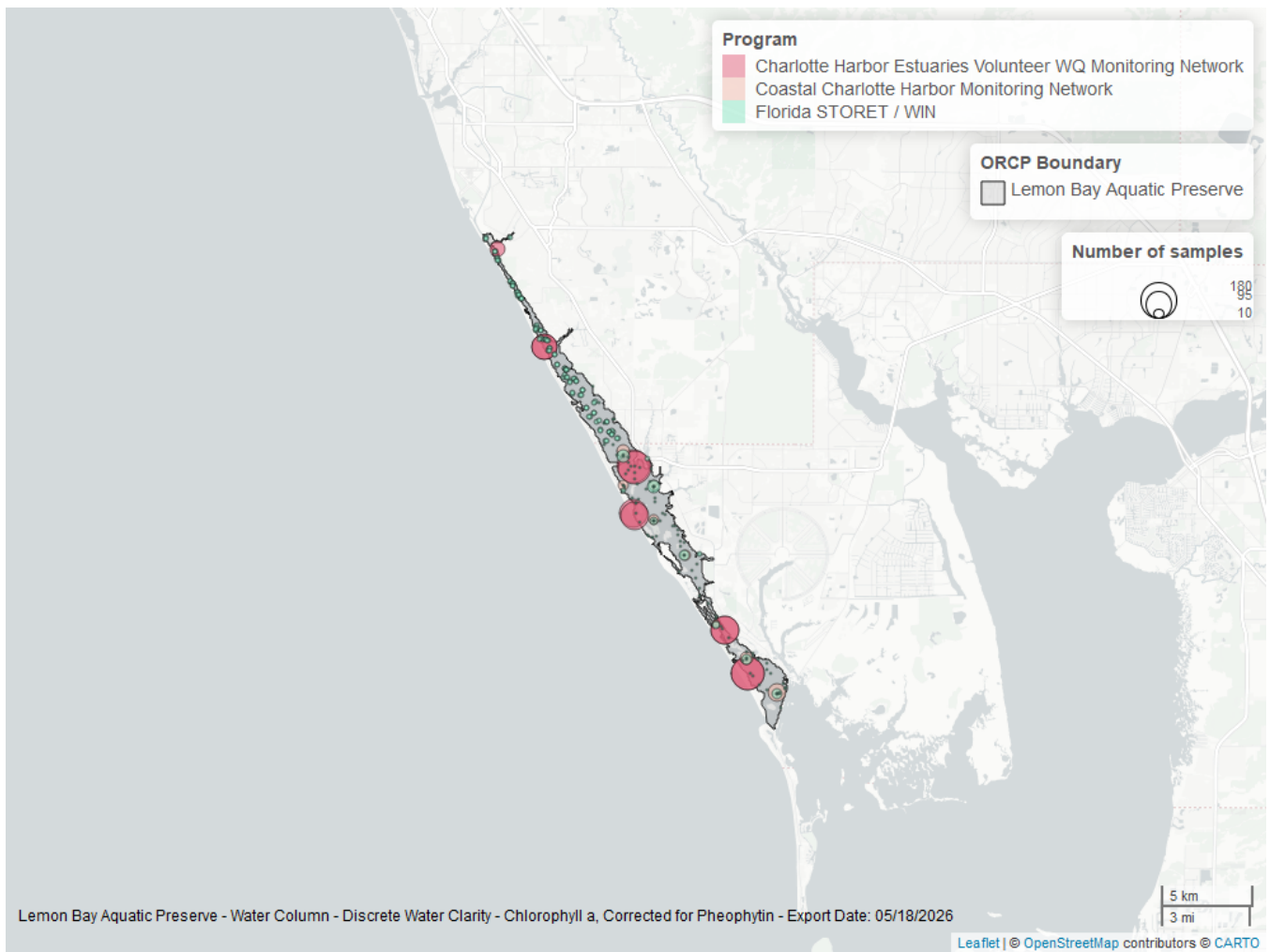


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 7: Programs contributing data for Chlorophyll a, Corrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
476	739	2008	2025
5002	390	1998	2025
513	179	2001	2009

Program names:

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network¹

5002 - Florida STORET / WIN²

513 - Coastal Charlotte Harbor Monitoring Network³

Chlorophyll a, Uncorrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

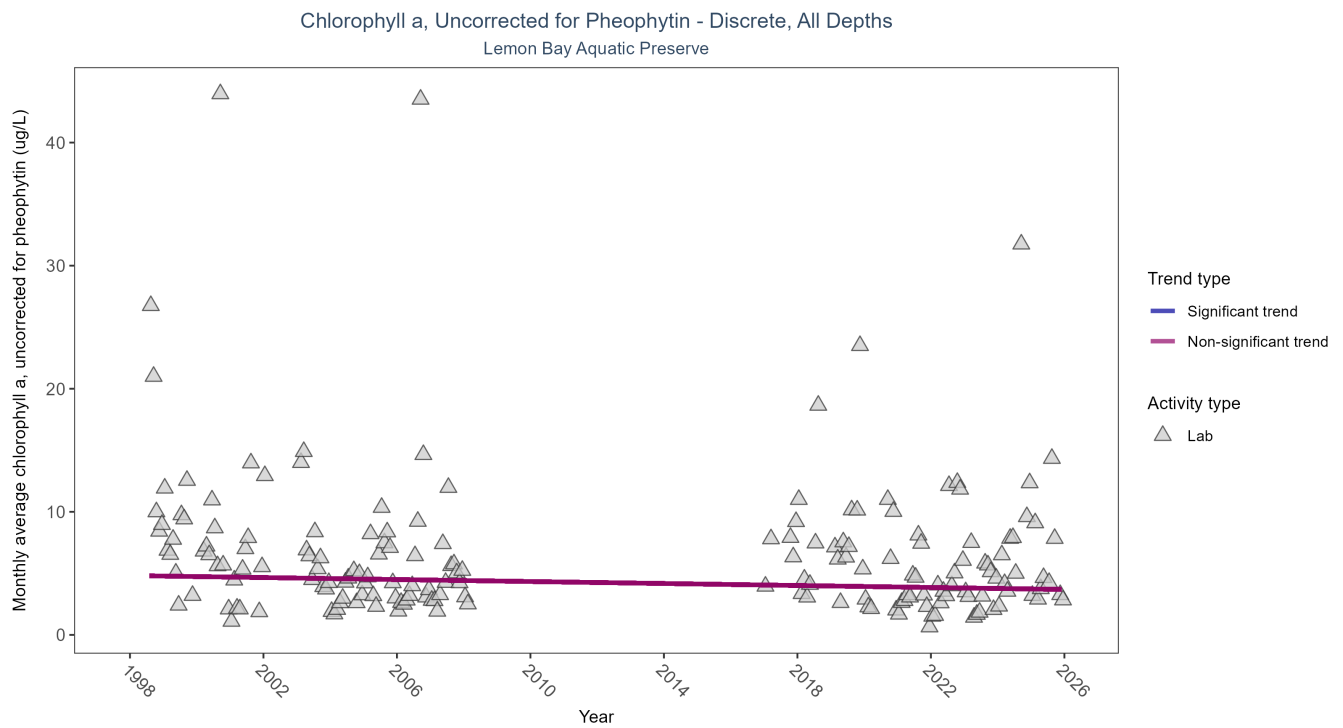


Figure 3: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 8: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	795	20	1998 - 2025	4.1	-0.093	4.8191	-0.0403	0.1066

Chlorophyll a, uncorrected for pheophytin, showed no detectable trend between 1998 and 2025.

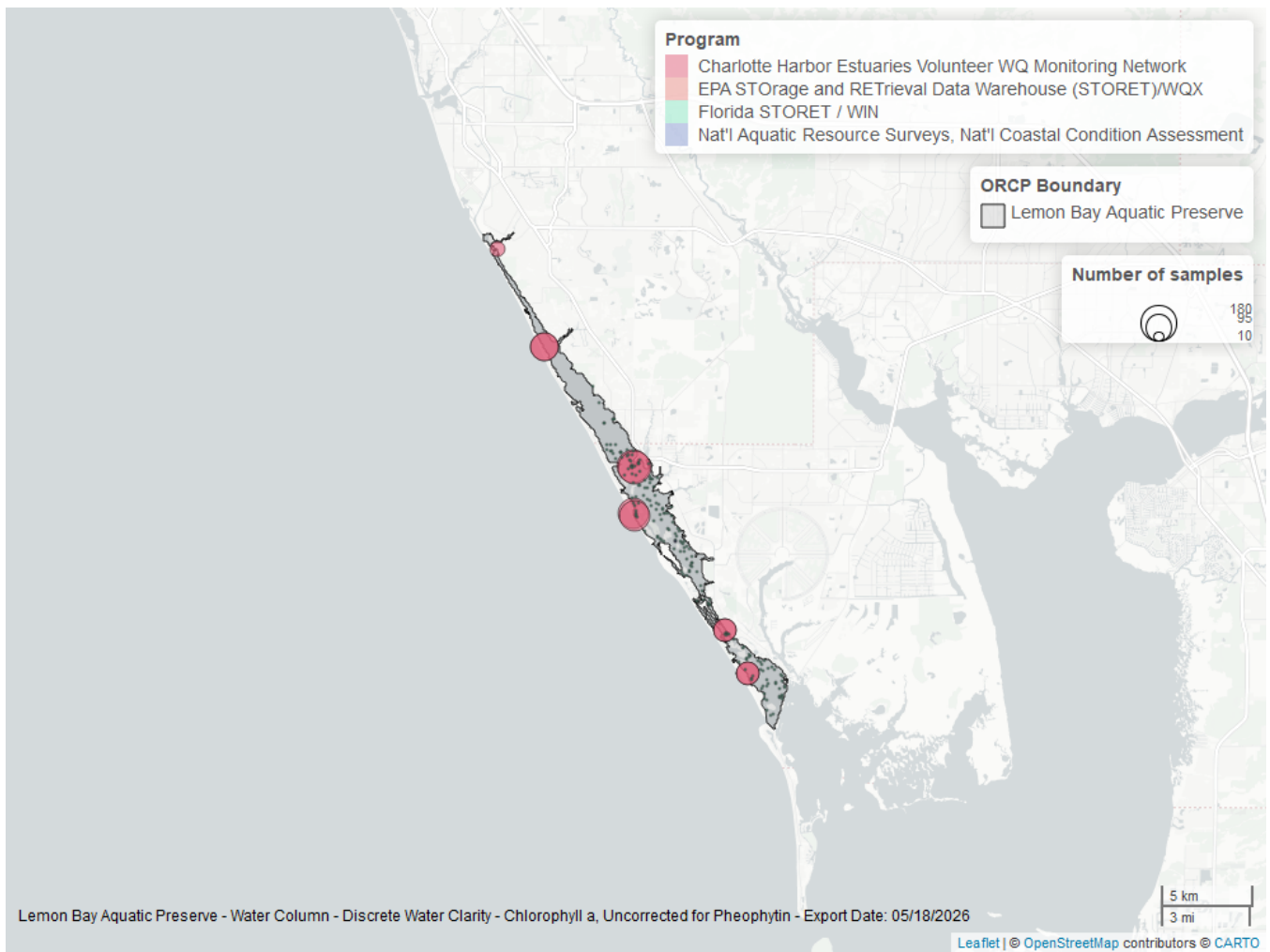


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 9: Programs contributing data for Chlorophyll a, Uncorrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
476	629	1998	2025
5002	178	2017	2023
103	2	2005	2005
118	1	2005	2005

Program names:

[476](#) - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network¹

5002 - Florida STORET / WIN²

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁴

118 - National Aquatic Resource Surveys, National Coastal Condition Assessment⁵

Colored Dissolved Organic Matter - Discrete

Seasonal Kendall-Tau Trend Analysis

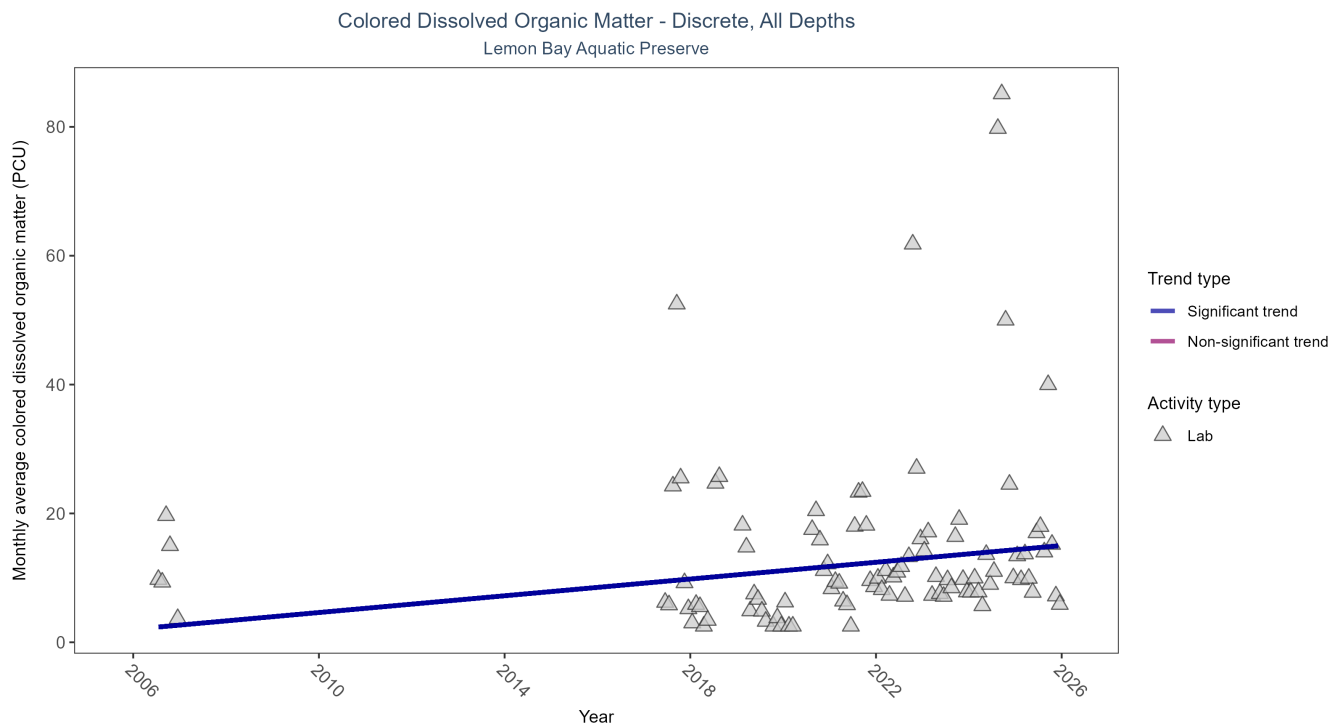


Figure 5: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 10: Seasonal Kendall-Tau Trend Analysis for Colored Dissolved Organic Matter

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	587	10	2006 - 2025	10	0.3076	2.0194	0.65	0.0007

Monthly average colored dissolved organic matter increased by 0.65 PCU per year, indicating a decrease in water clarity.

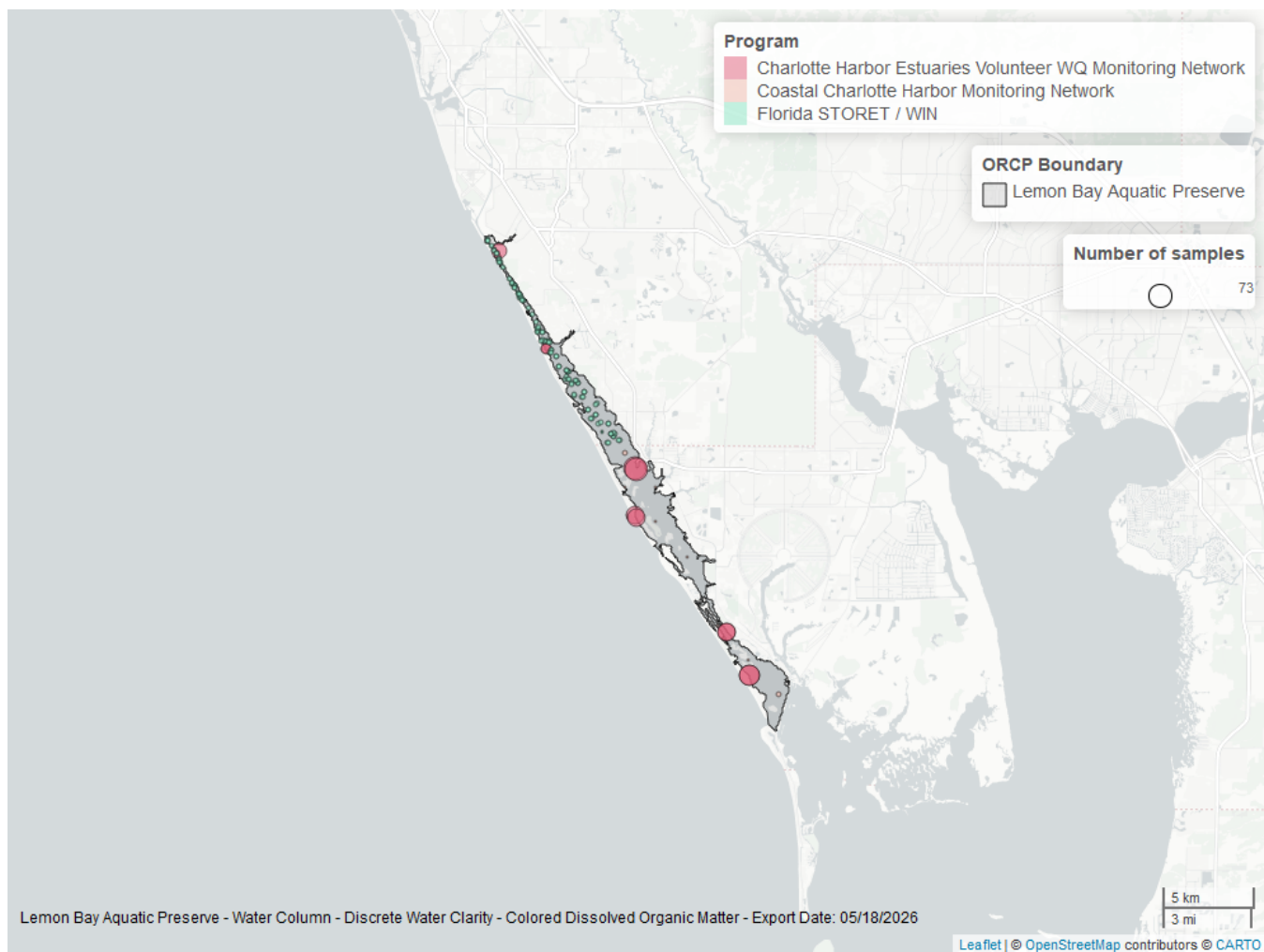


Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 11: Programs contributing data for Colored Dissolved Organic Matter

ProgramID	N_Data	YearMin	YearMax
476	291	2017	2025
5002	284	2020	2025
513	14	2006	2006

Program names:

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network¹
 5002 - Florida STORET / WIN²
 513 - Coastal Charlotte Harbor Monitoring Network³

Dissolved Oxygen - Discrete

Seasonal Kendall-Tau Trend Analysis

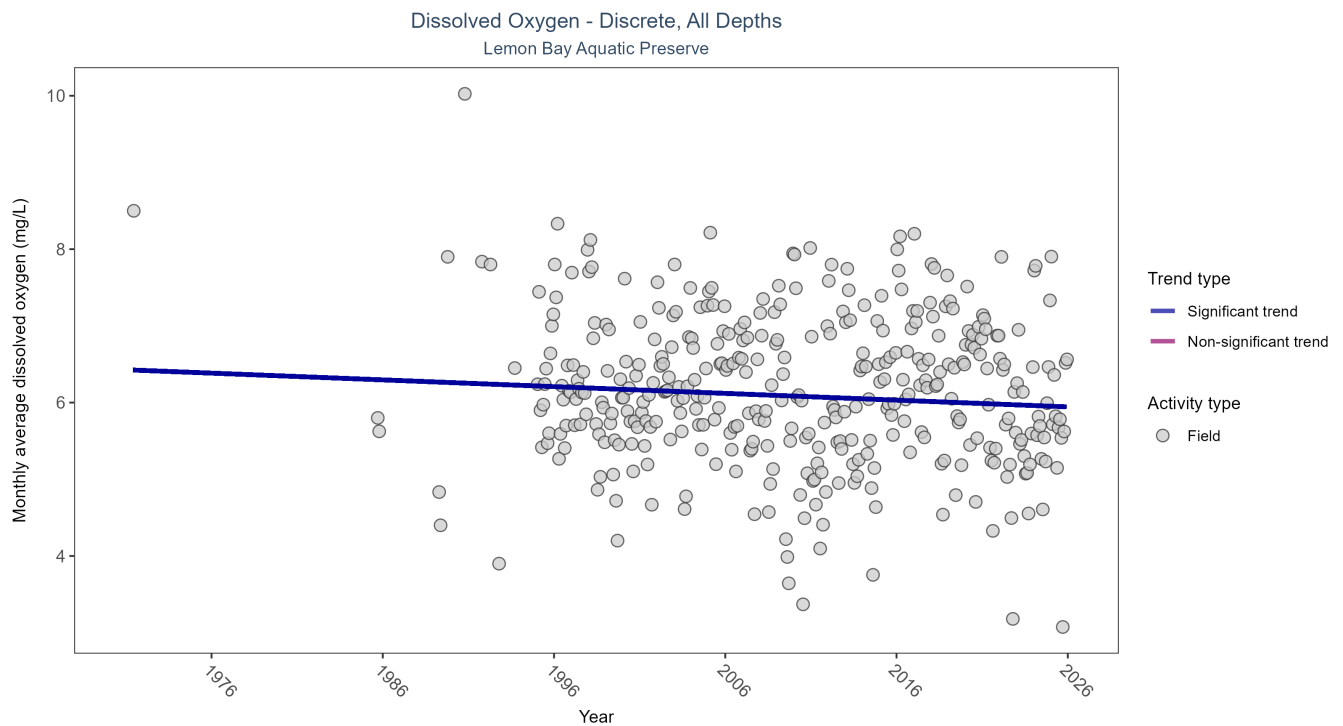


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	13794	38	1971 - 2025	6.1	-0.0661	6.4277	-0.0088	0.0496

Monthly average dissolved oxygen decreased by 0.01 mg/L per year.

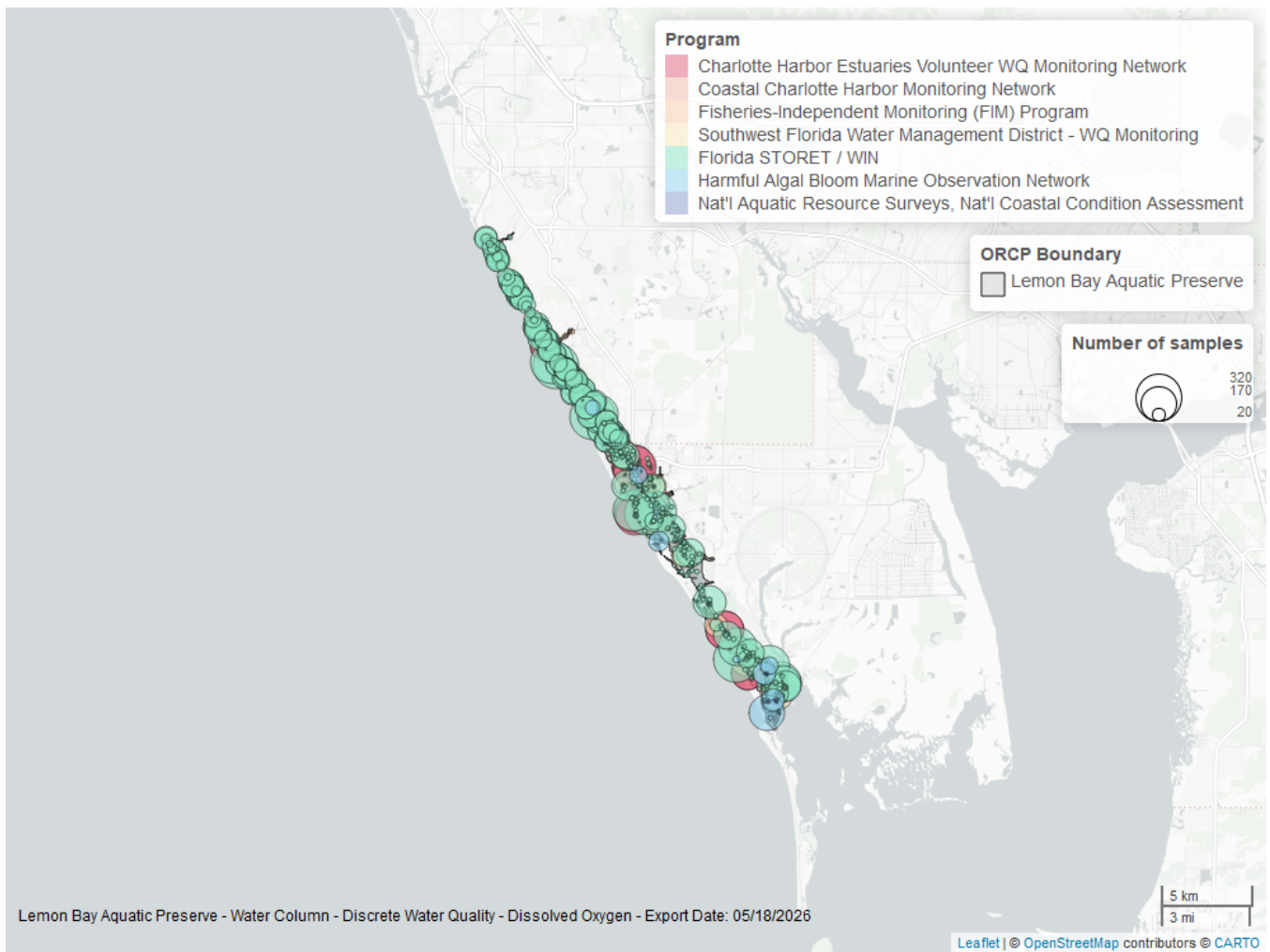


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 13: Programs contributing data for Dissolved Oxygen

ProgramID	N_Data	YearMin	YearMax
5002	10043	1995	2025
476	1050	1998	2025
69	917	1989	2024
479	865	2001	2016
95	647	1971	2018
513	407	2001	2009
118	5	2005	2020

Program names:

5002 - Florida STORET / WIN²

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network¹

69 - Fisheries-Independent Monitoring (FIM) Program⁶

479 - Southwest Florida Water Management District - Water Quality Monitoring⁷

95 - Harmful Algal Bloom Marine Observation Network⁸

Dissolved Oxygen Saturation - Discrete

Seasonal Kendall-Tau Trend Analysis

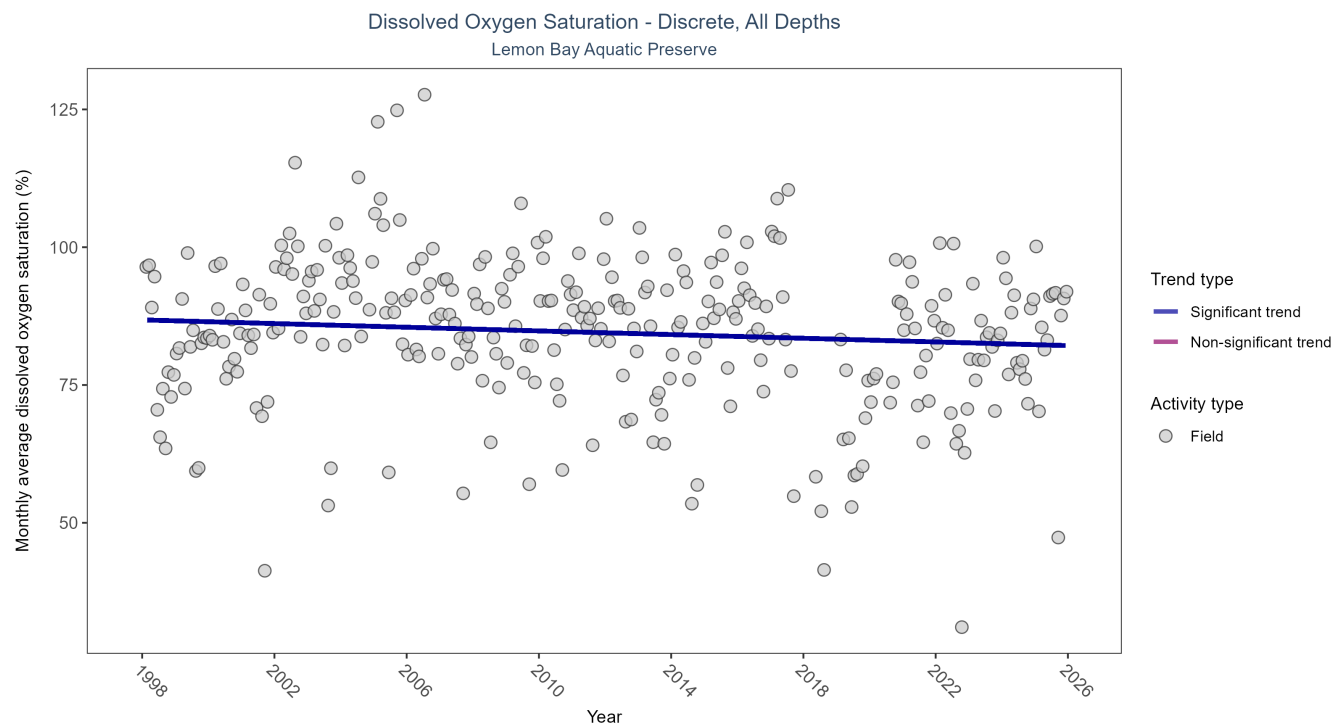


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 14: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen Saturation

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	4703	28	1998 - 2025	87	-0.0876	86.8133	-0.1667	0.0286

Monthly average dissolved oxygen saturation decreased by 0.17% per year.

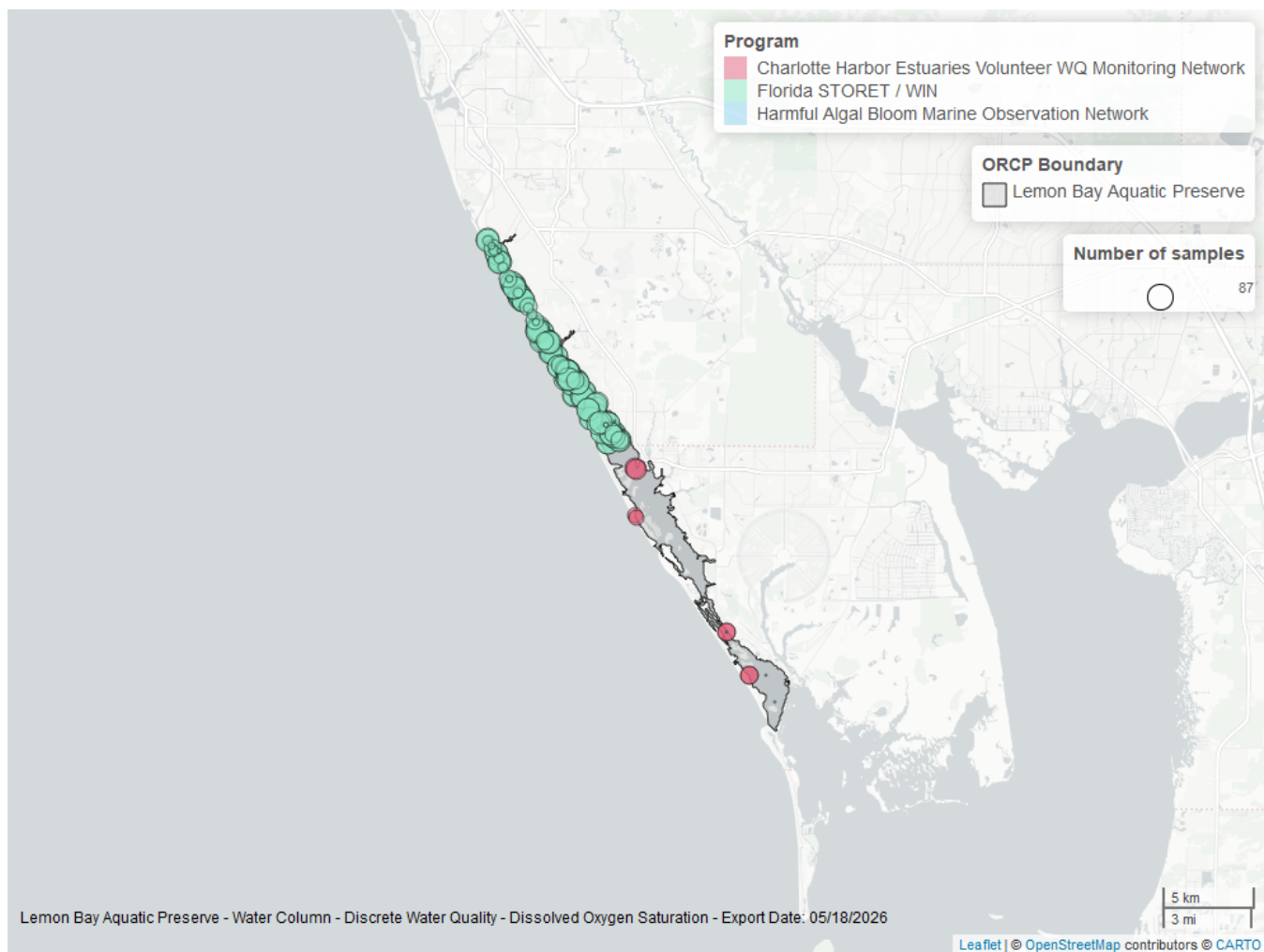


Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 15: Programs contributing data for Dissolved Oxygen Saturation

ProgramID	N_Data	YearMin	YearMax
5002	4508	1998	2025
476	247	2018	2025
95	4	2005	2018

Program names:

5002 - Florida STORET / WIN²

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network¹

95 - Harmful Algal Bloom Marine Observation Network⁸

pH - Discrete

Seasonal Kendall-Tau Trend Analysis

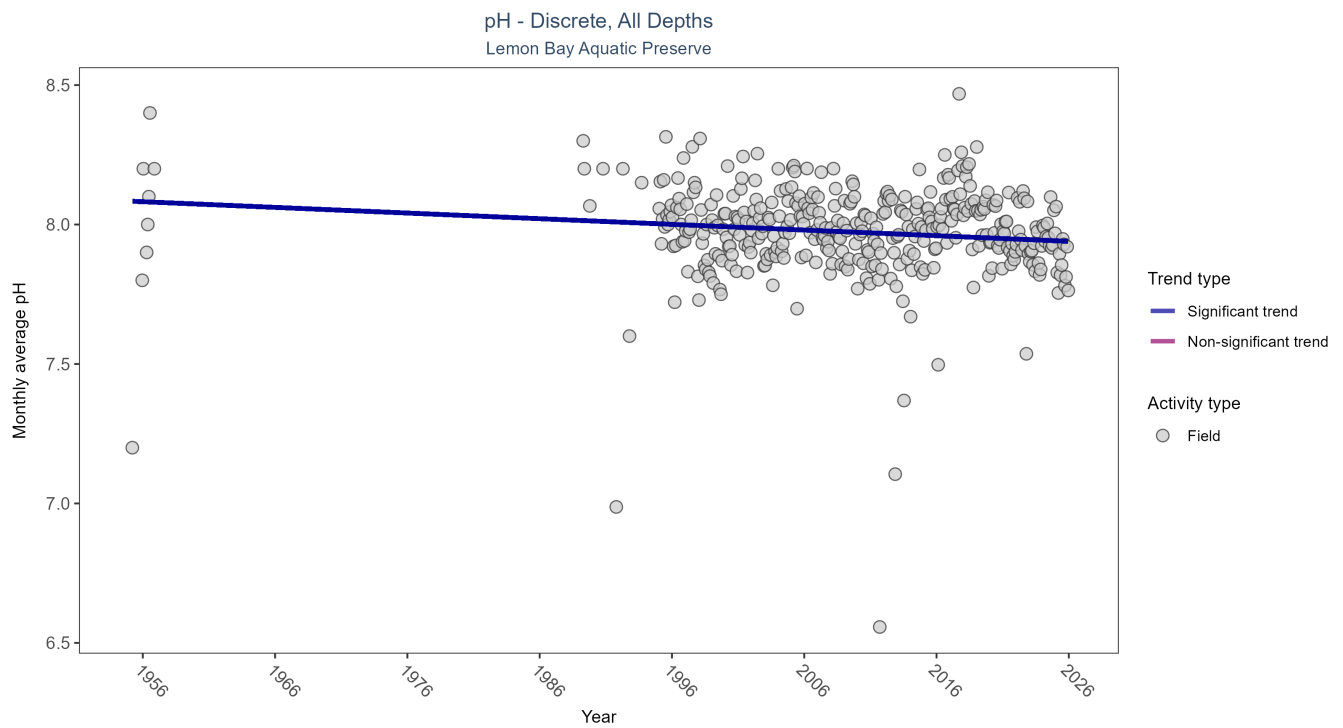


Figure 11: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 16: Seasonal Kendall-Tau Trend Analysis for pH

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	12226	38	1955 - 2025	8	-0.1088	8.0838	-0.002	0.002

Monthly average pH decreased by less than 0.01 pH units per year.

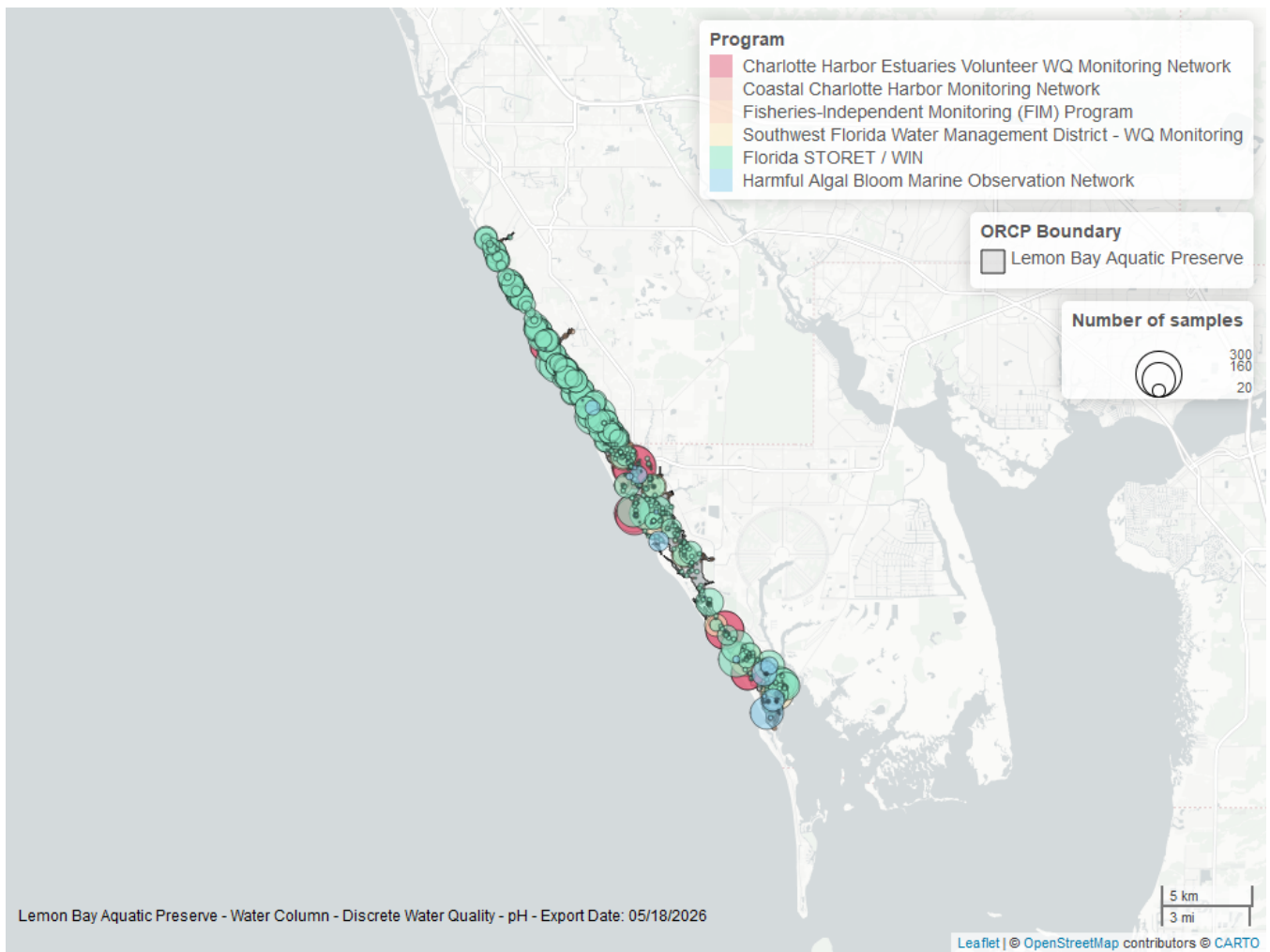


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 17: Programs contributing data for pH

ProgramID	N_Data	YearMin	YearMax
5002	8431	1995	2025
476	1066	1998	2025
69	915	1989	2024
479	869	2001	2016
95	625	1955	2018
513	407	2001	2009

Program names:

5002 - Florida STORET / WIN²

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network¹

69 - Fisheries-Independent Monitoring (FIM) Program⁶

479 - Southwest Florida Water Management District - Water Quality Monitoring⁷

95 - Harmful Algal Bloom Marine Observation Network⁸

513 - Coastal Charlotte Harbor Monitoring Network³

Salinity - Discrete

Seasonal Kendall-Tau Trend Analysis

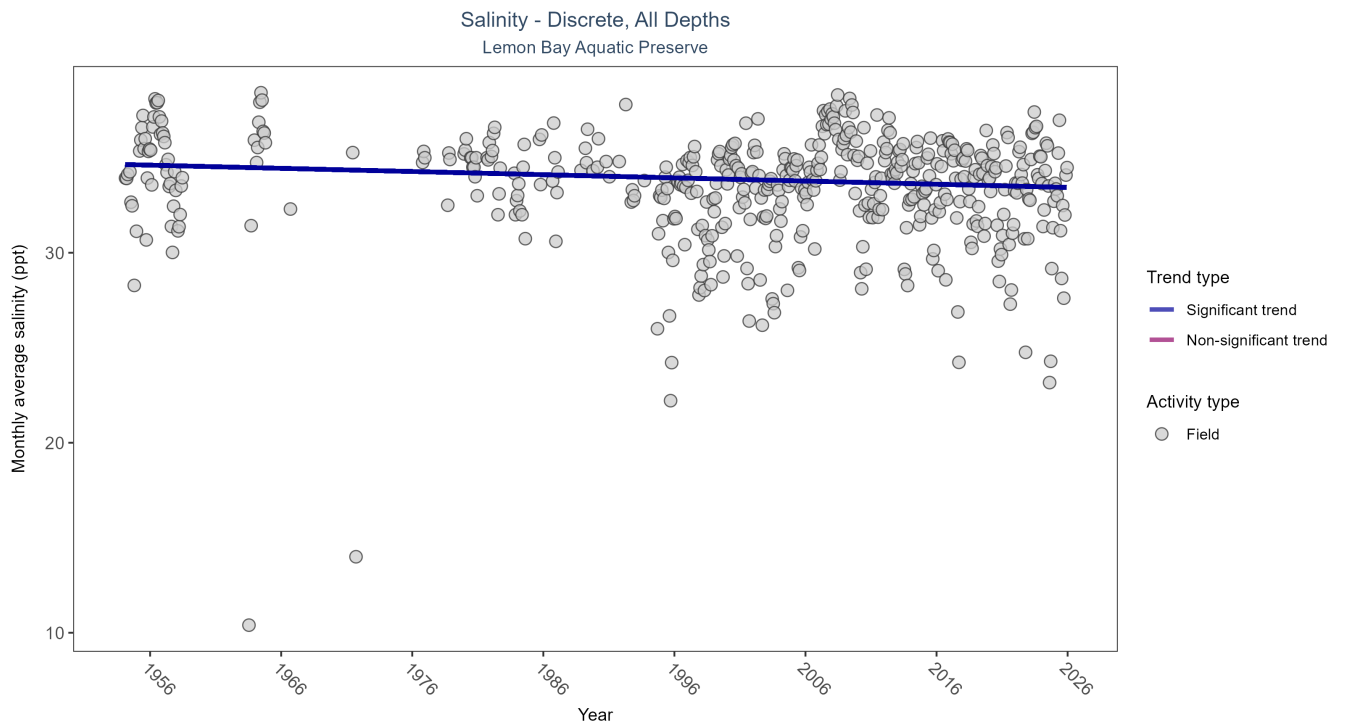


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 18: Seasonal Kendall-Tau Trend Analysis for Salinity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Decreasing trend	11486	58	1954 - 2025	34	-0.0891	34.6434	-0.0167	0.0024

Monthly average salinity decreased by 0.02 ppt per year.

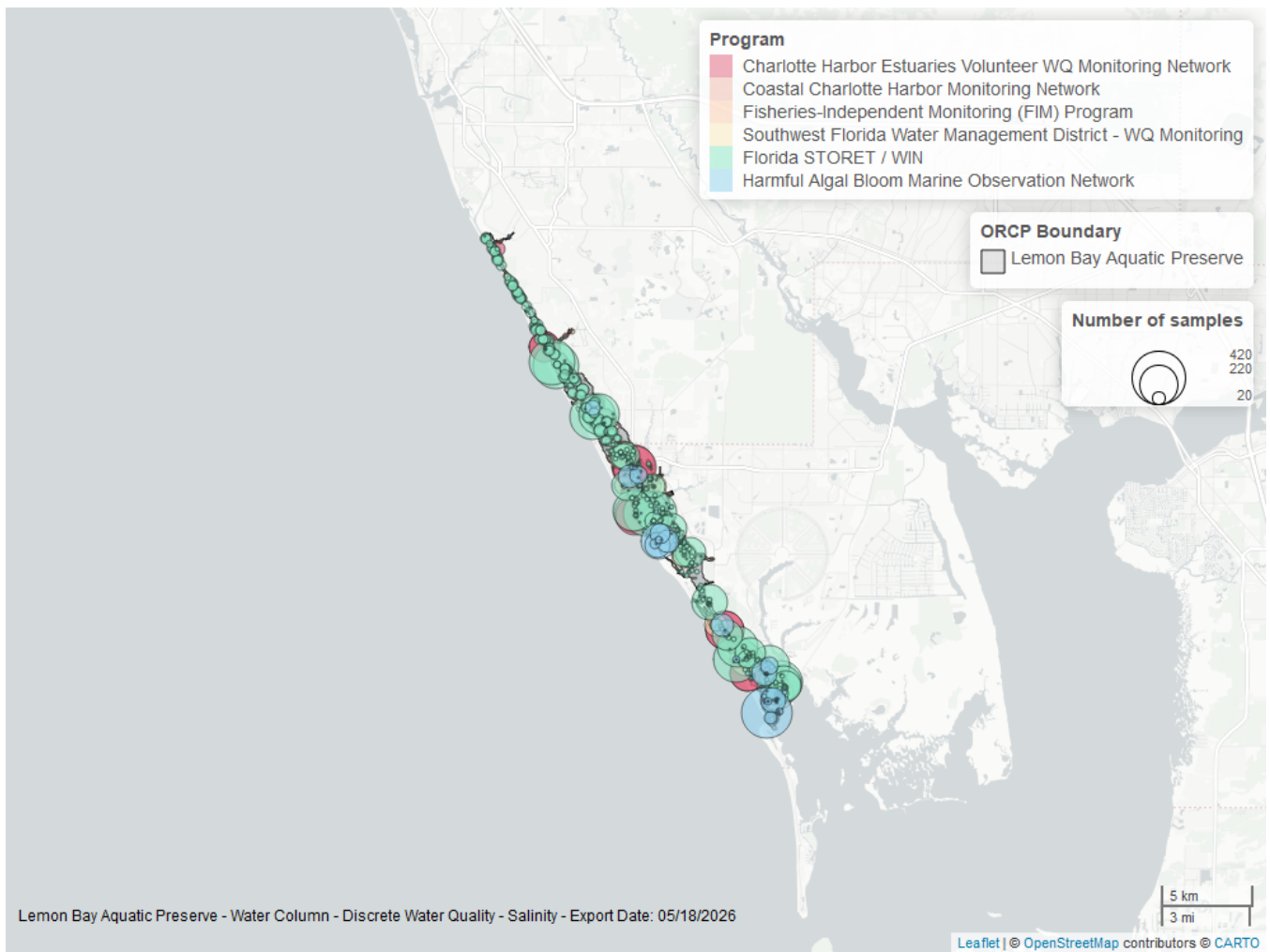


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 19: Programs contributing data for Salinity

ProgramID	N_Data	YearMin	YearMax
5002	6704	1995	2025
95	1521	1954	2018
476	1075	1998	2025
69	928	1989	2024
479	870	2001	2016
513	407	2001	2009

Program names:

5002 - Florida STORET / WIN²

95 - Harmful Algal Bloom Marine Observation Network⁸

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network¹

69 - Fisheries-Independent Monitoring (FIM) Program⁶

479 - Southwest Florida Water Management District - Water Quality Monitoring⁷

513 - Coastal Charlotte Harbor Monitoring Network³

Total Nitrogen - Discrete

Total Nitrogen Calculation:

The logic for calculated Total Nitrogen was provided by Kevin O'Donnell and colleagues at FDEP (with the help of Jay Silvanima, Watershed Monitoring Section). The following logic is used, in this order, based on the availability of specific nitrogen components.

- 1) $TN = TKN + NO3O2$;
- 2) $TN = TKN + NO3 + NO2$;
- 3) $TN = ORGN + NH4 + NO3O2$;
- 4) $TN = ORGN + NH4 + NO2 + NO3$;
- 5) $TN = TKN + NO3$;
- 6) $TN = ORGN + NH4 + NO3$;

Additional Information:

- Rules for use of sample fraction:
 - Florida Department of Environmental Protection (FDEP) report that if both “Total” and “Dissolved” components are reported, only “Total” is used. If the total is not reported, then the dissolved components are used as a best available replacement.
 - Total nitrogen calculations are done using nitrogen components with the same sample fraction, nitrogen components with mixed total/dissolved sample fractions are not used. In other words, total nitrogen can be calculated when TKN and NO3O2 are both total sample fractions, or when both are dissolved sample fractions. *Future calculations of total nitrogen values may be based on components with mixed sample fractions.*
- Values inserted into data:
 - ParameterName = “Total Nitrogen”
 - SEACAR_QAQCFlagCode = “1Q”
 - SEACAR_QAQC_Description = “SEACAR Calculated”

Seasonal Kendall-Tau Trend Analysis

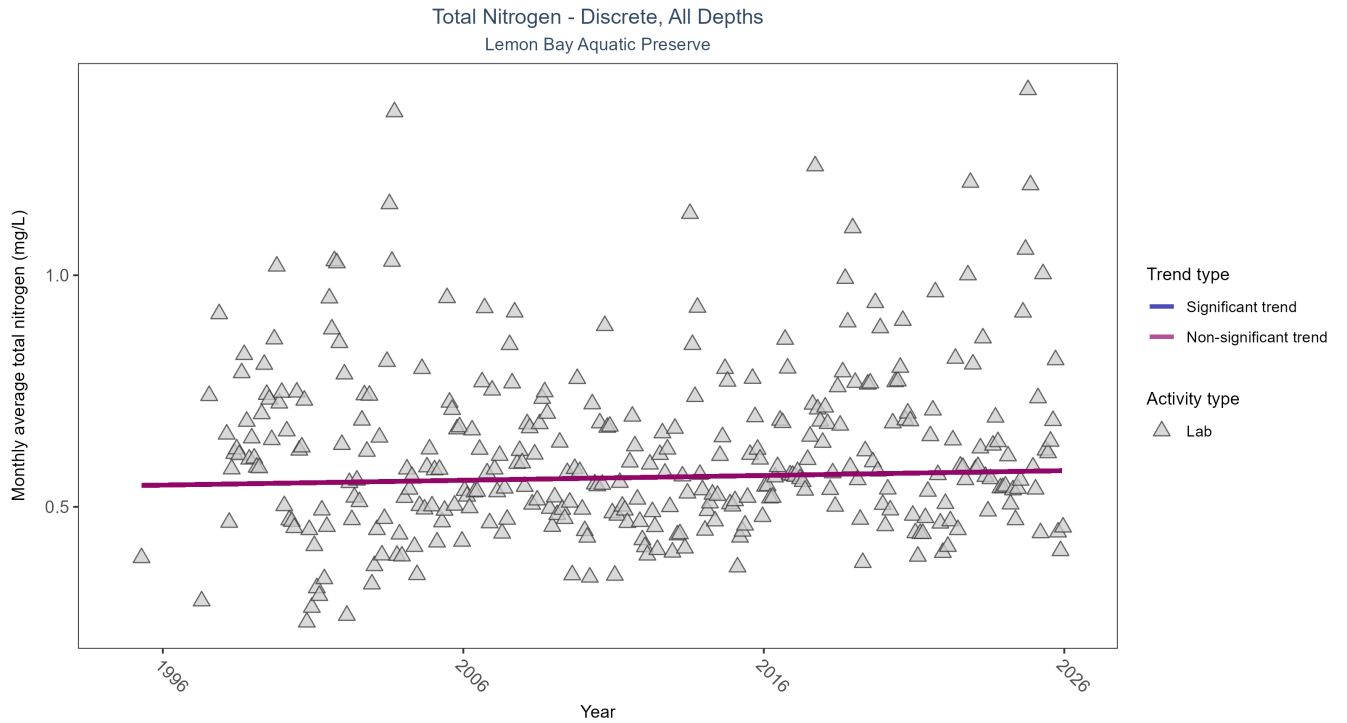


Figure 15: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 20: Seasonal Kendall-Tau Trend Analysis for Total Nitrogen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	3125	30	1995 - 2025	0.555	0.0473	0.5458	0.001	0.2139

Total nitrogen showed no detectable trend between 1995 and 2025.

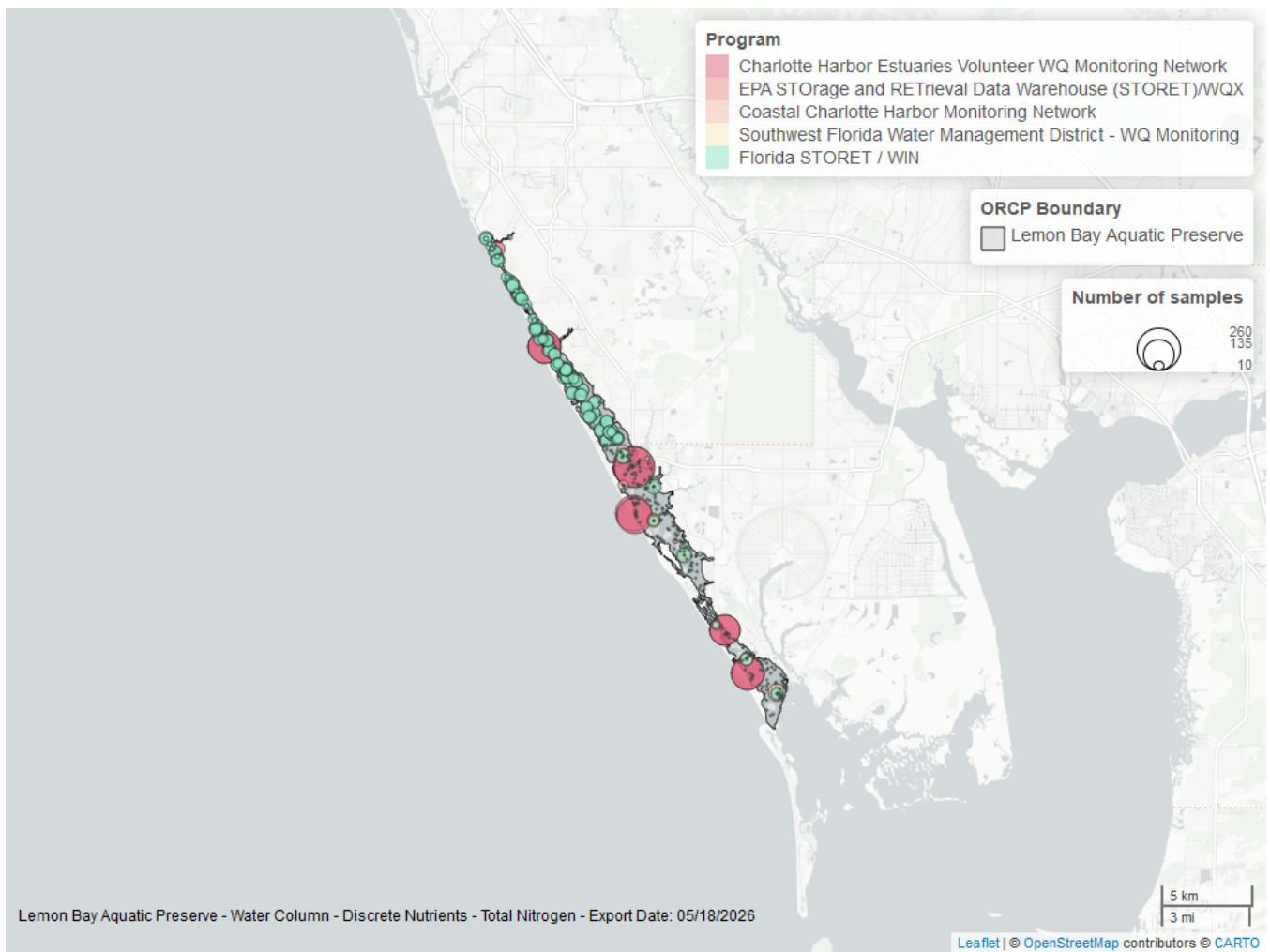


Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 21: Programs contributing data for Total Nitrogen

ProgramID	N_Data	YearMin	YearMax
5002	1810	1995	2025
476	981	1998	2025
513	181	2001	2009
479	152	2007	2016
103	4	2005	2005

Program names:

5002 - Florida STORET / WIN²

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network¹

513 - Coastal Charlotte Harbor Monitoring Network³

479 - Southwest Florida Water Management District - Water Quality Monitoring⁷

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁴

Total Phosphorus - Discrete

Seasonal Kendall-Tau Trend Analysis

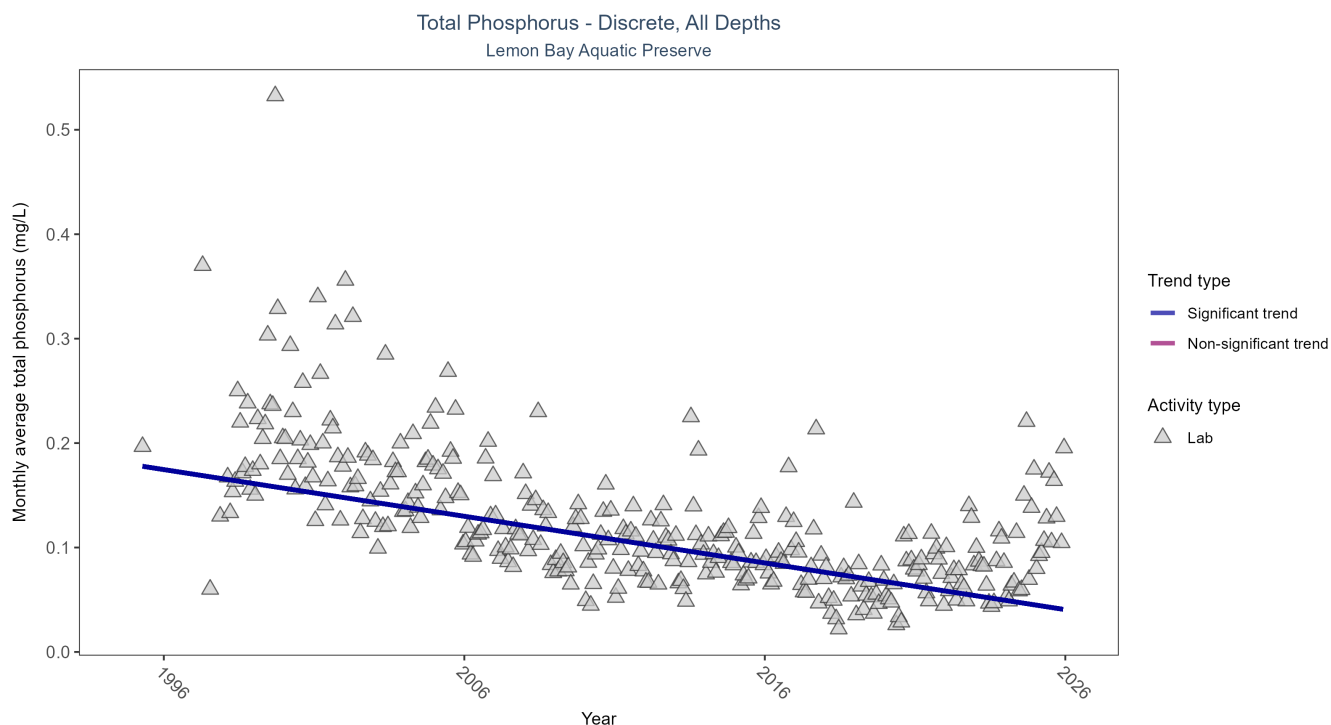


Figure 17: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 22: Seasonal Kendall-Tau Trend Analysis for Total Phosphorus

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	3172	30	1995 - 2025	0.085	-0.5088	0.1791	-0.0045	0

Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

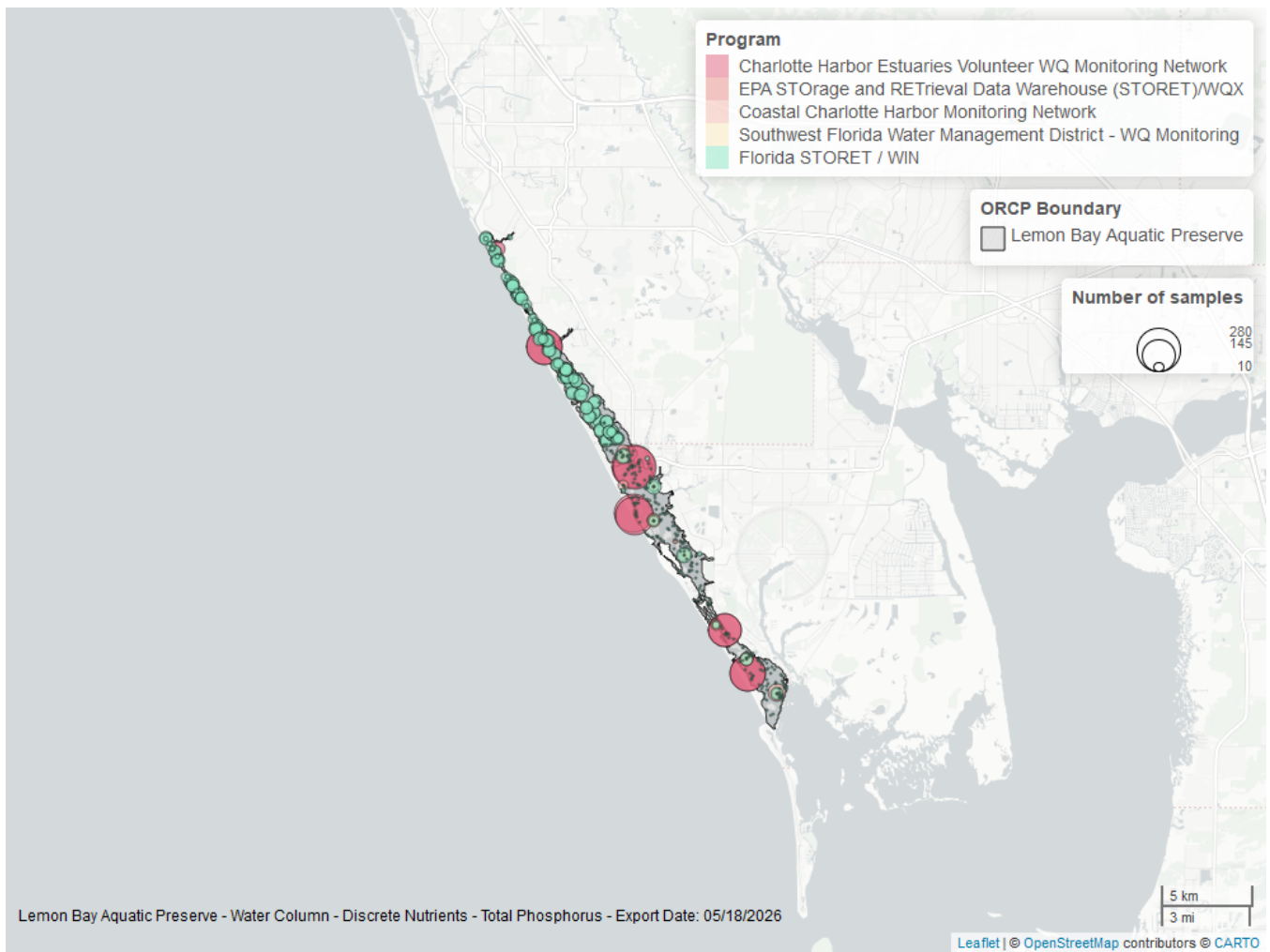


Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 23: Programs contributing data for Total Phosphorus

ProgramID	N_Data	YearMin	YearMax
5002	1816	1995	2025
476	1093	1998	2025
513	184	2001	2009
479	152	2007	2016
103	4	2005	2005

Program names:

5002 - Florida STORET / WIN²

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network¹

513 - Coastal Charlotte Harbor Monitoring Network³

479 - Southwest Florida Water Management District - Water Quality Monitoring⁷

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁴

Turbidity - Discrete

Seasonal Kendall-Tau Trend Analysis

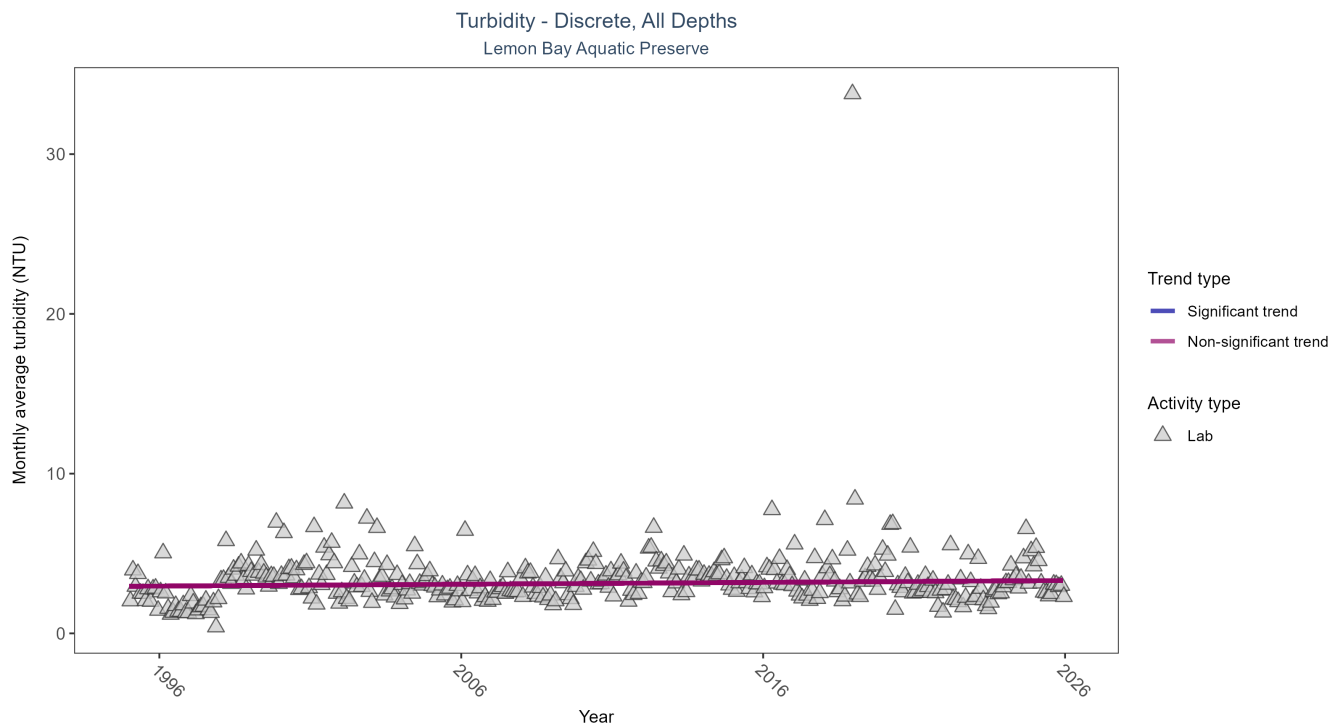


Figure 19: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 24: Seasonal Kendall-Tau Trend Analysis for Turbidity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	6205	31	1995 - 2025	2.7	0.0591	2.948	0.0116	0.1065

Turbidity showed no detectable trend between 1995 and 2025.

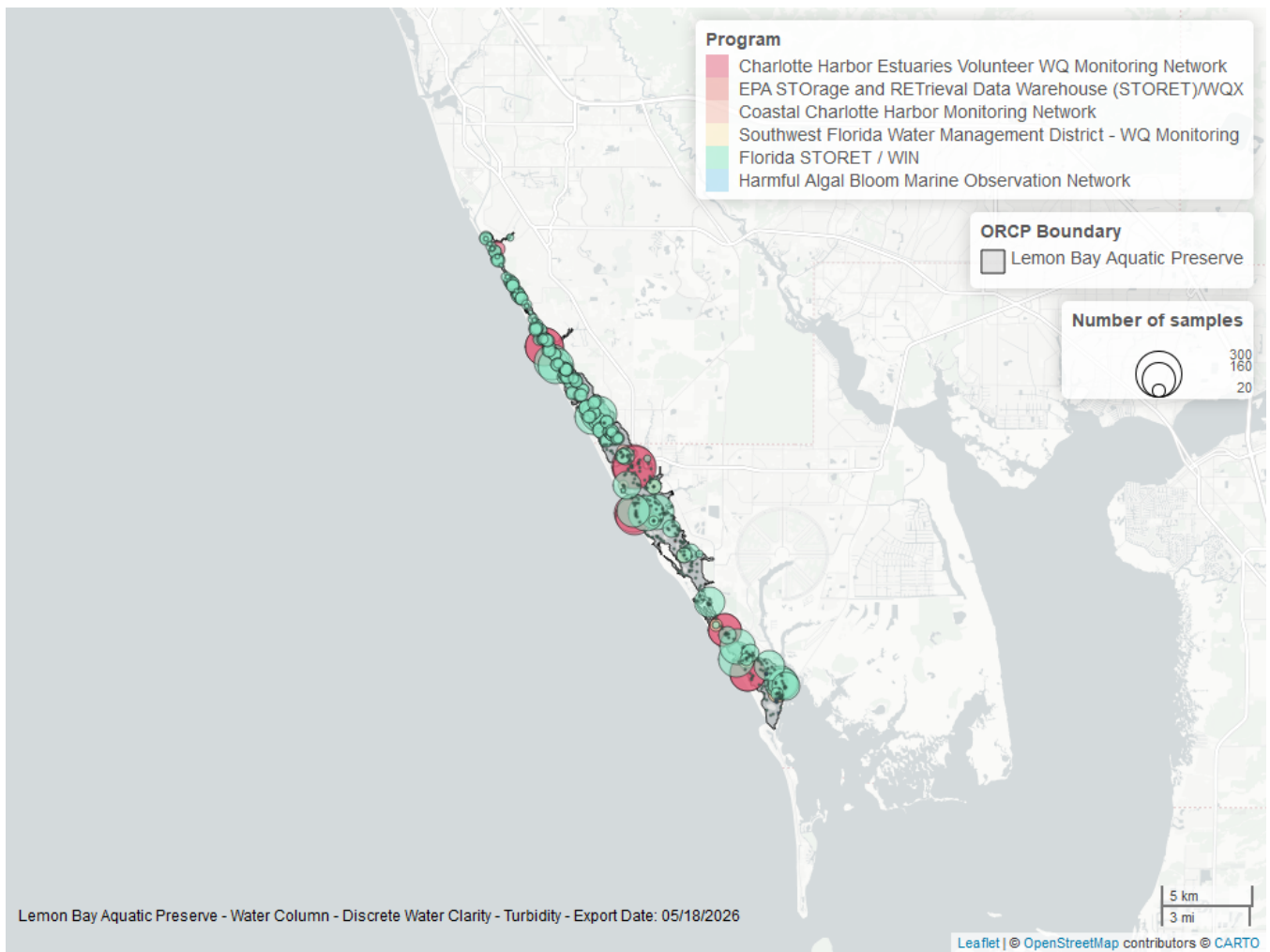


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 25: Programs contributing data for Turbidity

ProgramID	N_Data	YearMin	YearMax
5002	4714	1995	2025
476	1132	1998	2025
479	278	2001	2016
513	185	2001	2009
95	14	2003	2004
103	2	2005	2005

Program names:

5002 - Florida STORET / WIN²

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network¹

479 - Southwest Florida Water Management District - Water Quality Monitoring⁷

513 - Coastal Charlotte Harbor Monitoring Network³

95 - Harmful Algal Bloom Marine Observation Network⁸

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁴

Water Temperature - Discrete

Seasonal Kendall-Tau Trend Analysis

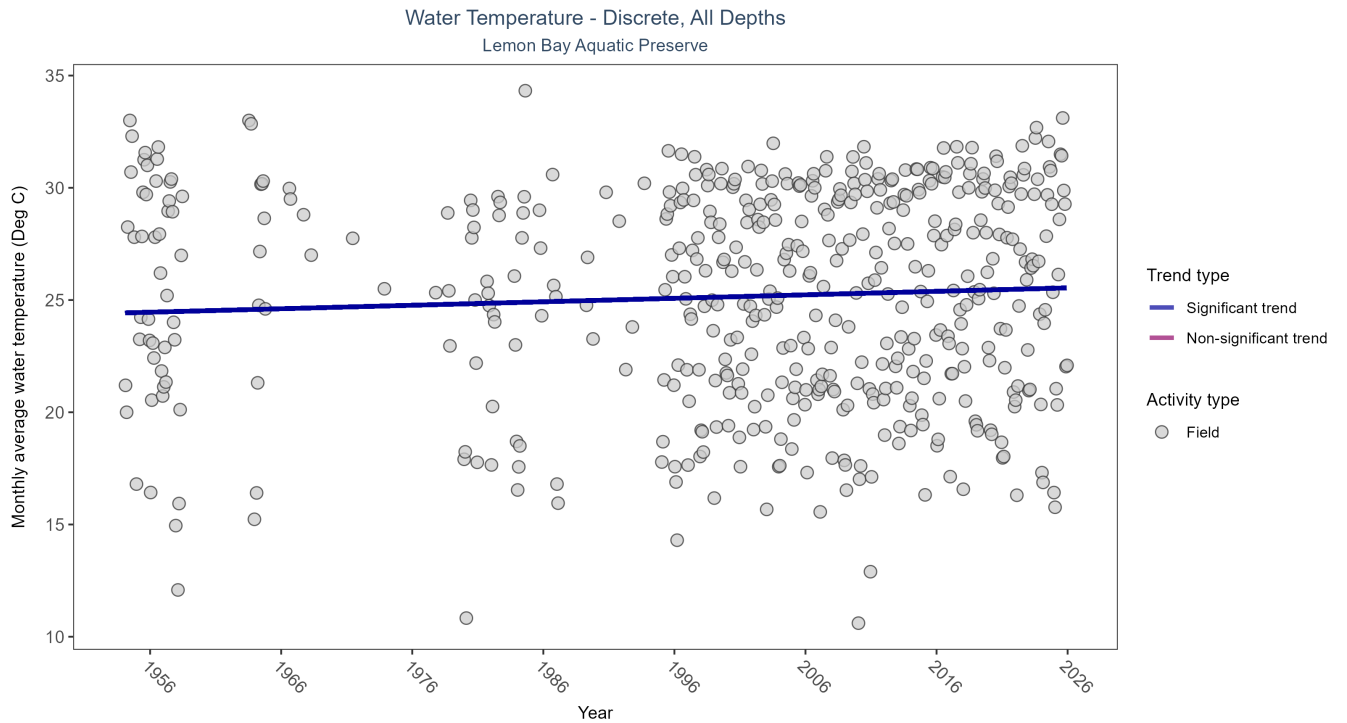


Figure 21: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 26: Seasonal Kendall-Tau Trend Analysis for Water Temperature

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	14987	59	1954 - 2025	26.4	0.1243	24.426	0.0155	0

Monthly average water temperature increased by 0.02°C per year.

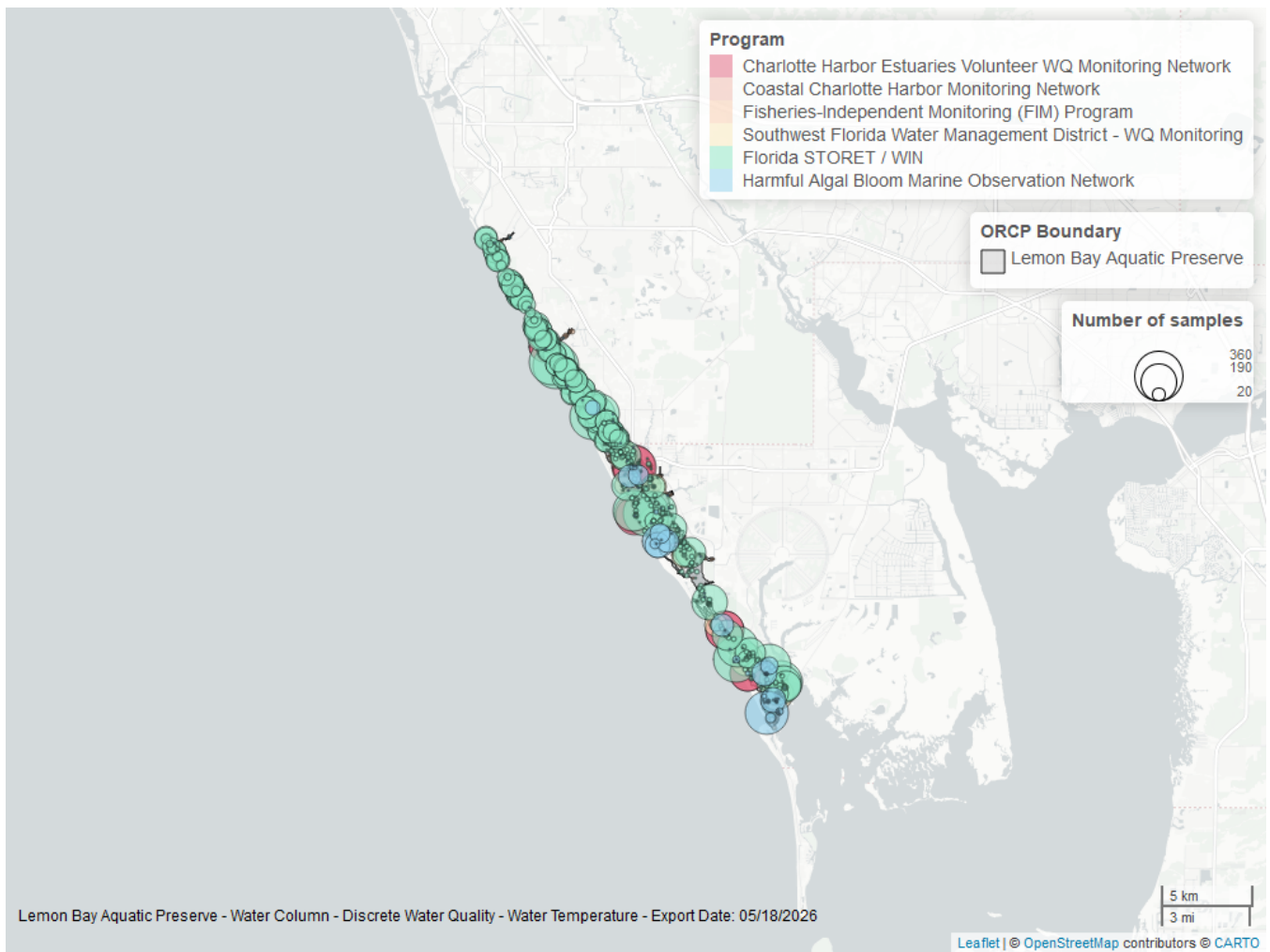


Figure 22: Map showing location of discrete water quality sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 27: Programs contributing data for Water Temperature

ProgramID	N_Data	YearMin	YearMax
5002	10384	1995	2025
95	1330	1954	2018
476	1082	1998	2025
69	933	1989	2024
479	865	2001	2016
513	407	2001	2009

Program names:

5002 - Florida STORET / WIN²

95 - Harmful Algal Bloom Marine Observation Network⁸

476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network¹

69 - Fisheries-Independent Monitoring (FIM) Program⁶

479 - Southwest Florida Water Management District - Water Quality Monitoring⁷

513 - Coastal Charlotte Harbor Monitoring Network³

Submerged Aquatic Vegetation

The data file used is: **All_SAV_Parameters-2026-Jun-04.txt**

Submerged aquatic vegetation (SAV) refers to plants and plant-like macroalgae species that live entirely underwater. The two primary categories of SAV inhabiting Florida estuaries are *benthic macroalgae* and *seagrasses*. They often grow together in dense beds or meadows that carpet the seafloor. *Macroalgae* include multicellular species of green, red and brown algae that often live attached to the substrate by a holdfast. They tend to grow quickly and can tolerate relatively high nutrient levels, making them a threat to seagrasses and other benthic habitats in areas with poor water quality. In contrast, *seagrasses* are grass-like, vascular, flowering plants that are attached to the seafloor by extensive root systems. *Seagrasses* occur throughout the coastal areas of Florida, including protected bays and lagoons as well as deeper offshore waters on the continental shelf. *Seagrasses* have taken advantage of the broad, shallow shelf and clear water to produce two of the most extensive seagrass beds anywhere in continental North America.

Parameters

Percent Cover measures the fraction of an area of seafloor that is covered by SAV, usually estimated by evaluating multiple small areas of seafloor. Percent cover is often estimated for total SAV, individual types of vegetation (seagrass, attached algae, drift algae) and individual species.

Frequency of Occurrence was calculated as the number of times a taxon was observed in a year divided by the number of sampling events, multiplied by 100. Analysis is conducted at the quadrat level and is inclusive of all quadrats (i.e., quadrats evaluated using Braun-Blanquet, modified Braun-Blanquet, and percent cover.”

Species

Turtle grass (*Thalassia testudinum*) is the largest of the Florida seagrasses, with longer, thicker blades and deeper root structures than any of the other seagrasses. It is considered a climax seagrass species.

Shoal grass (*Halodule wrightii*) is an early colonizer of vegetated areas and usually grows in water too shallow for other species except *widgeon grass*. It can often tolerate larger salinity ranges than other seagrass species. *Shoal grass* is characterized by thin, flat blades, that are narrower than *turtle grass* blades.

Manatee grass (*Syringodium filiforme*) is easily recognizable because its leaves are thin and cylindrical instead of the flat, ribbon-like form shared by many other seagrass species. The leaves can grow up to half a meter in length. *Manatee grass* is usually found in mixed seagrass beds or small, dense monospecific patches.

Widgeon grass (*Ruppia maritima*) grows in both fresh and salt water and is widely distributed throughout Florida's estuaries in less saline areas, particularly in inlets along the east coast. This species resembles *shoal grass* in certain environments but can be identified by the pointed tips of its leaves.

Three species of *Halophila* spp. are found in Florida - **Star grass** (*Halophila engelmannii*), **Paddle grass** (*Halophila decipiens*), and **Johnson's seagrass** (*Halophila johnsonii*). These are smaller, more fragile seagrasses than other Florida species and are considered ephemeral. They grow along a single long rhizome, with short blades. These species are not well-studied, although surveys are underway to define their ecological roles.

Notes

Star grass, *Paddle grass*, and *Johnson's seagrass* will be grouped together and listed as **Halophila spp.** in the following managed areas. This is because several surveys did not specify to the species level:

- Banana River Aquatic Preserve
- Indian River-Malabar to Vero Beach Aquatic Preserve
- Indian River-Vero Beach to Ft. Pierce Aquatic Preserve
- Jensen Beach to Jupiter Inlet Aquatic Preserve
- Loxahatchee River-Lake Worth Creek Aquatic Preserve
- Mosquito Lagoon Aquatic Preserve

- Biscayne Bay Aquatic Preserve
- Florida Keys National Marine Sanctuary

Lemon Bay Aquatic Preserve
SAV Percent Cover - Sample Locations

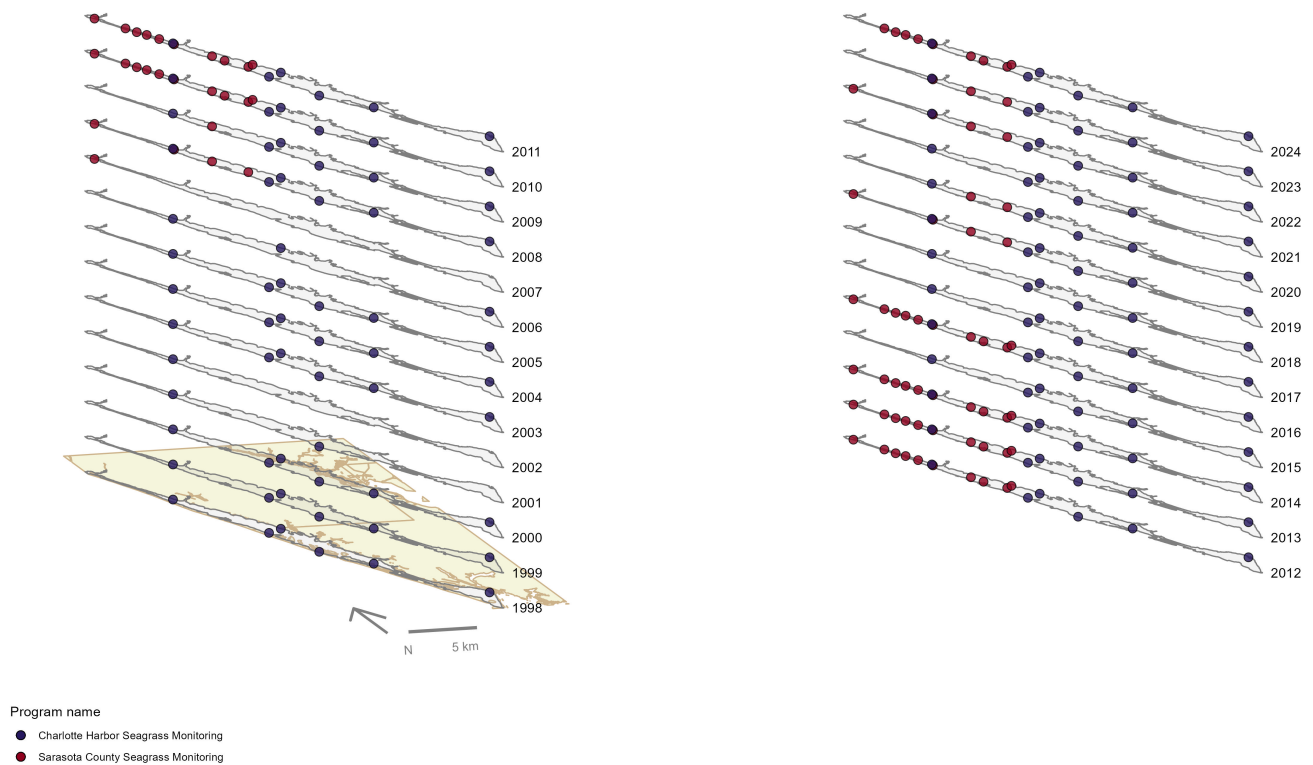


Figure 23: Maps showing the temporal scope of SAV sampling sites within the boundaries of *Lemon Bay Aquatic Preserve* by program name.

Click [here](#) to view spatio-temporal plots on GitHub.

Sampling locations by Program:

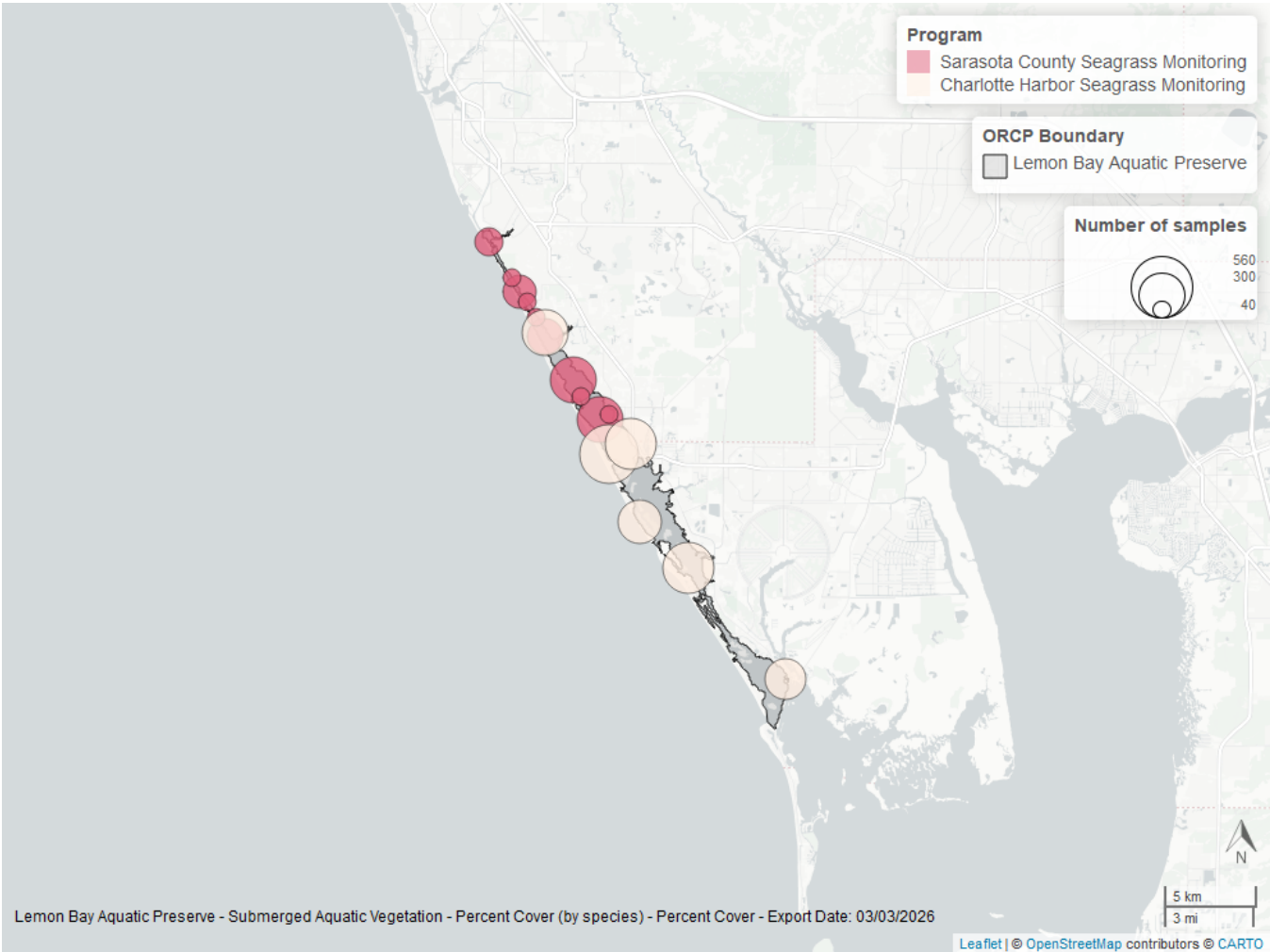


Figure 24: Map showing SAV sampling sites within the boundaries of *Lemon Bay Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Table 28: Program Information for Submerged Aquatic Vegetation

<i>ProgramID</i>	<i>N-Data</i>	<i>YearMin</i>	<i>YearMax</i>	<i>method</i>	<i>Sample Locations</i>
570	2209	1998	2024	Braun Blanquet	6
568	1397	2007	2024	Percent Cover	10

Program names:

- 568 - Sarasota County Seagrass Monitoring⁹
- [570](#) - Charlotte Harbor Seagrass Monitoring¹⁰

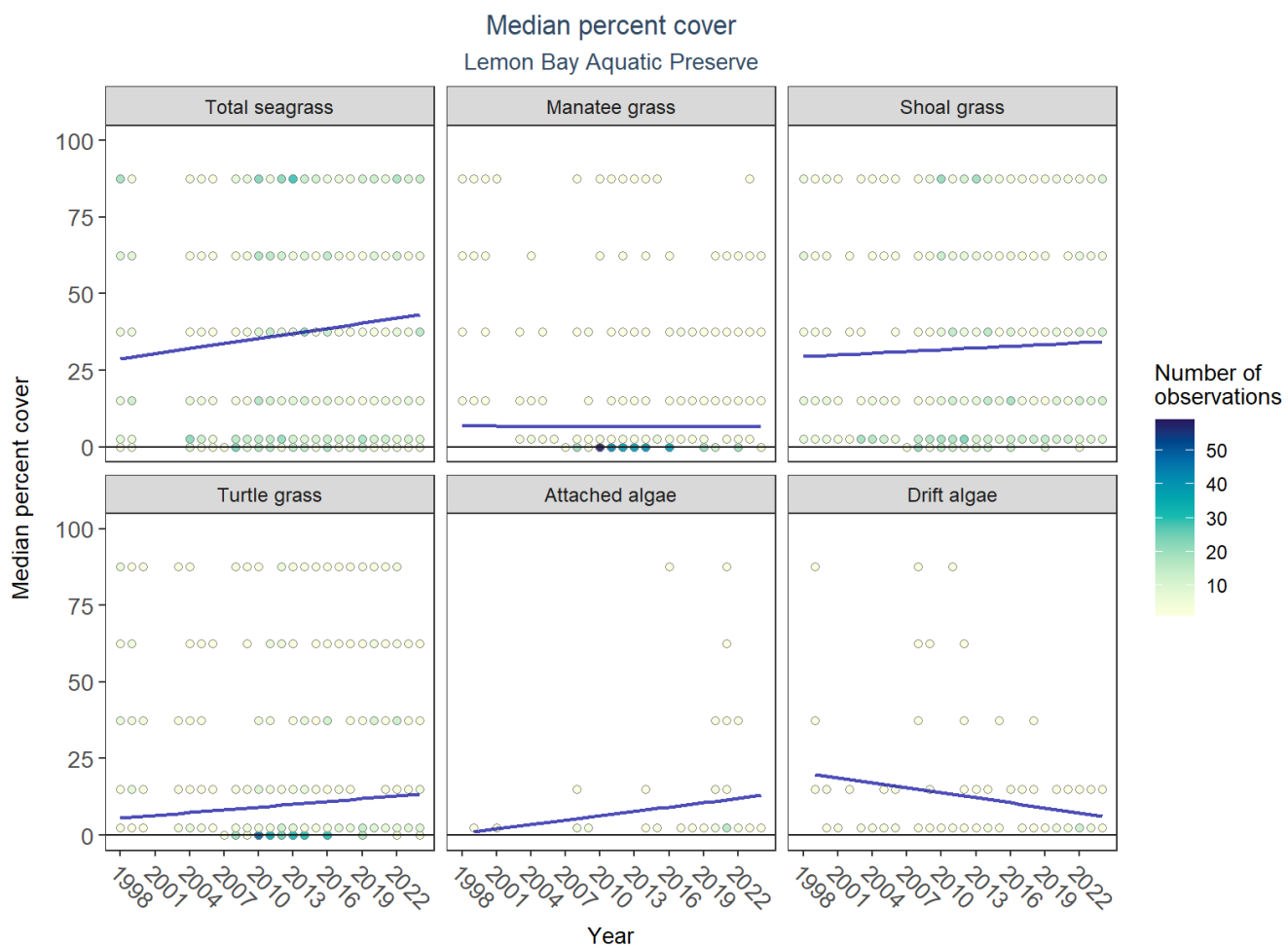


Figure 25: Scatter plots of median percent cover of submerged aquatic vegetation over time by group. Plots for time series that included five or more years of observations show the estimated trend as a blue line.

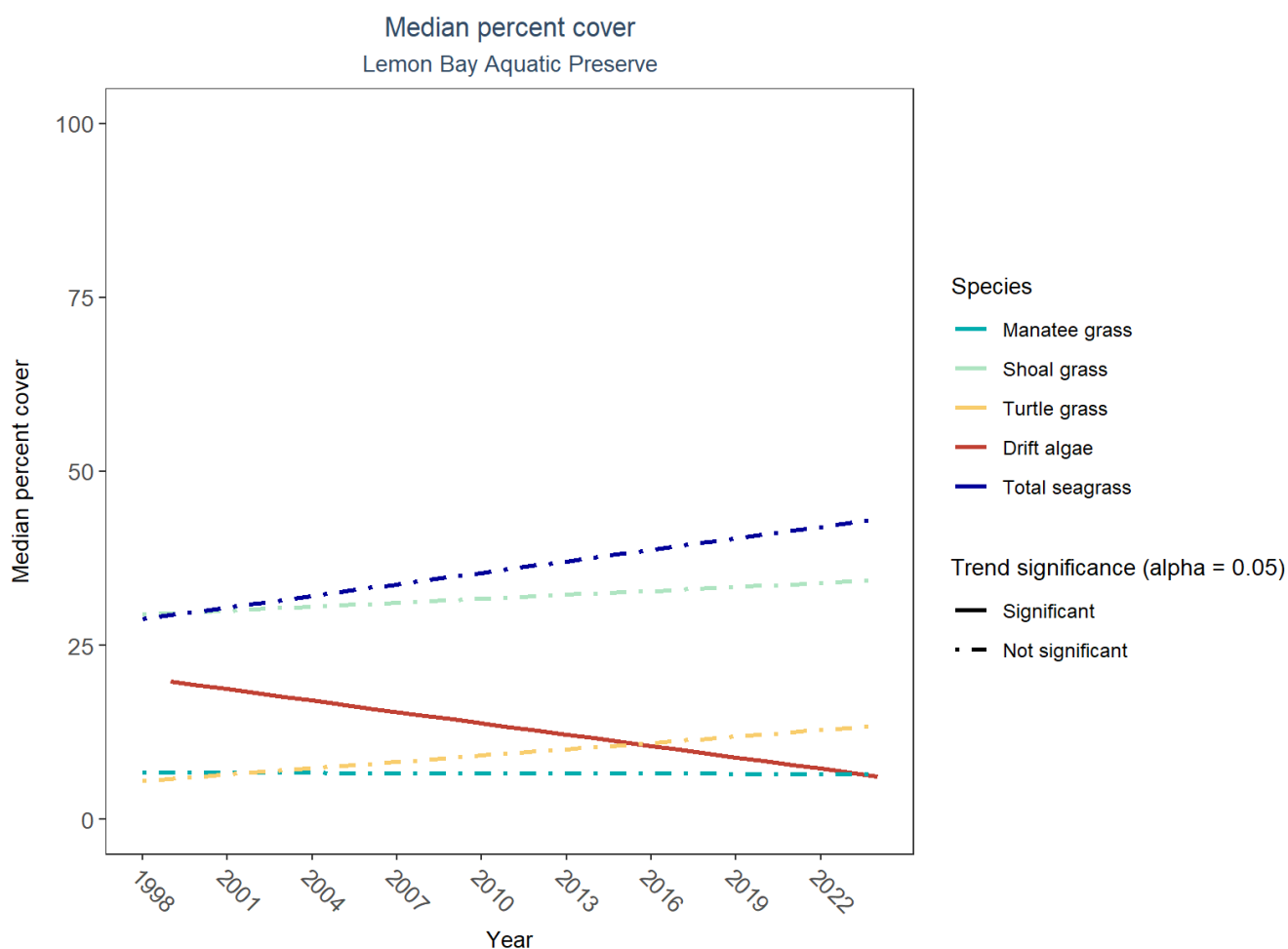


Figure 26: Trend lines of median percent cover of submerged aquatic vegetation over time for species that had five or more years of observations. Line type represents significance (solid) or non-significance (dashed) of the trends.

Table 29: Percent Cover Trend Analysis for Lemon Bay Aquatic Preserve

<i>Species</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>LME Intercept</i>	<i>LME Slope</i>	<i>P</i>
Attached algae	No detectable trend	1999 - 2024	-1.171794	0.4709814	0.2600356
Drift algae	Decreasing trend	1999 - 2024	22.530993	-0.5442751	0.0227362
Shoal grass	No detectable trend	1998 - 2024	28.685714	0.1887737	0.5983209
Manatee grass	No detectable trend	1998 - 2024	6.783818	-0.0086610	0.9626978
Turtle grass	No detectable trend	1998 - 2024	4.315207	0.3038590	0.4537040
Total seagrass	No detectable trend	1998 - 2024	26.641958	0.5494435	0.2467879

An annual decrease in percent cover was observed for drift algae (-0.54%). Total seagrass, manatee grass, shoal grass, turtle grass, and attached algae showed no detectable change in percent cover.

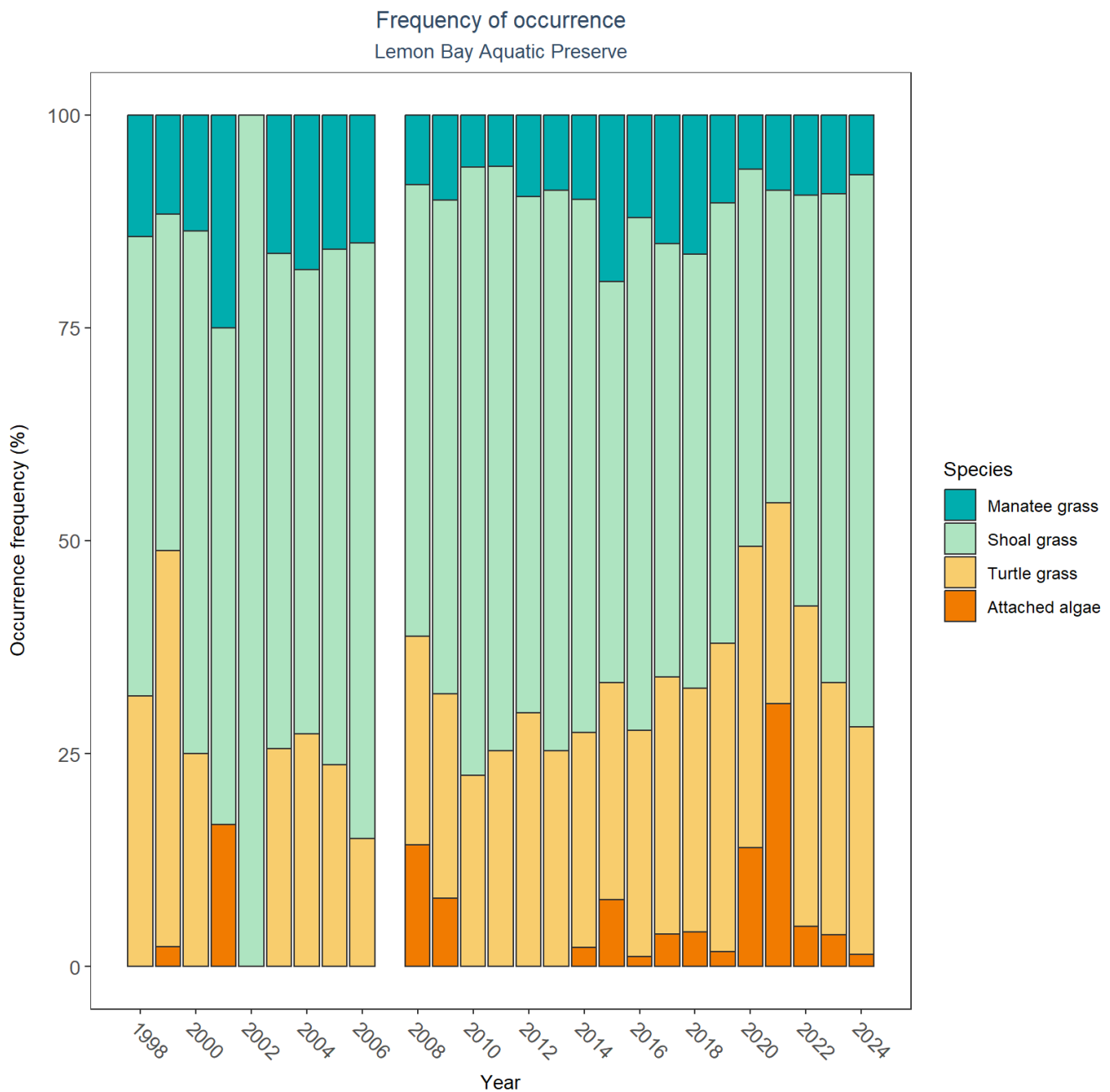


Figure 27: Bar plot of submerged aquatic vegetation species occurrence frequency over time by group.

SAV Water Column Analysis

The following parameters are available for Lemon Bay Aquatic Preserve within the SAV_WC_Report:

- Colored Dissolved Organic Matter
- Chlorophyll a
- Dissolved Oxygen
- Dissolved Oxygen Saturation
- pH
- Salinity

- Secchi Depth
- Water Temperature
- Total Nitrogen
- Total Suspended Solids
- Turbidity

Access the reports here: [DRAFT_SAV_WC_Report.pdf](#)

Nekton

The data file used is: **All_NEKTON_Parameters-2026-Jun-04.txt**

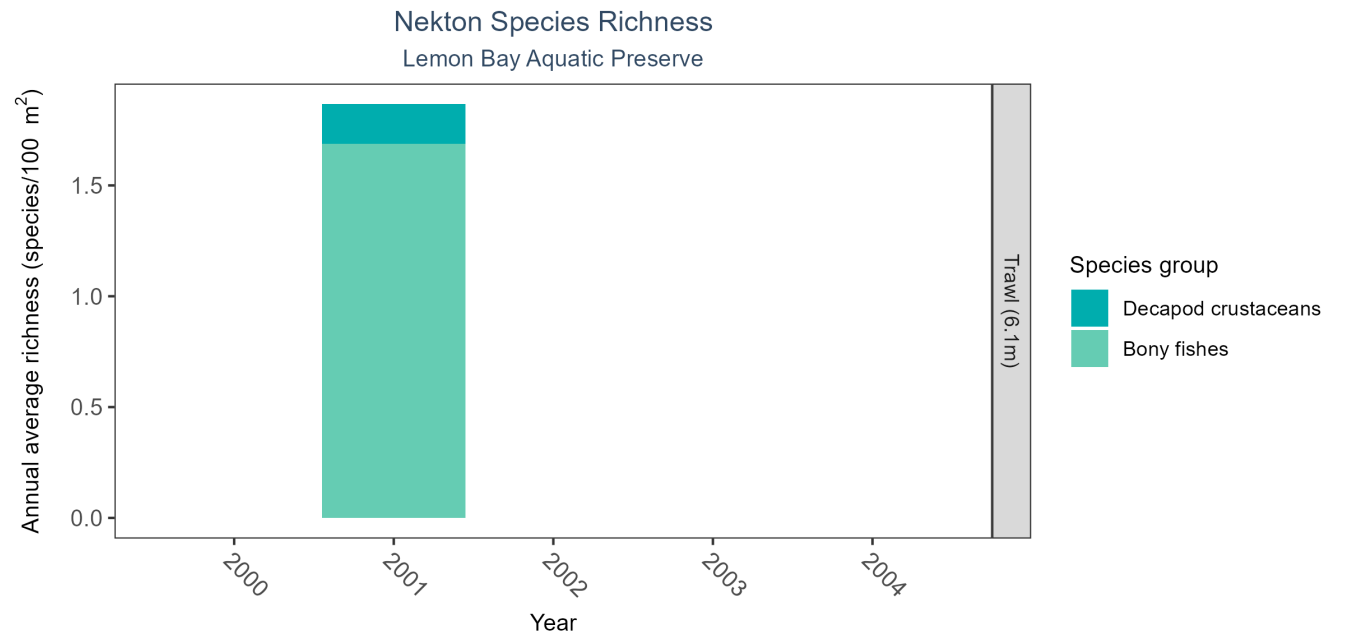


Figure 28: Bar graph(s) of annual average nekton richness over time for species groups occurring in at least 1% of samples. The bar colors represent species groups including bony fishes, cartilaginous fishes, decapod crustaceans (e.g., shrimps, crabs, and lobsters), and cephalopods (e.g., squid). Gear types and sizes are indicated in the panel label.

Table 30: Nekton Species Richness

<i>Gear Type</i>	<i>No. of Samples</i>	<i>No. of Years with Data</i>	<i>Period of Record</i>	<i>Median No. of Taxa</i>	<i>Mean No. of Taxa</i>
Trawl (6.1 m)	6	1	2001 - 2001	0.74	0.93

The median annual number of taxa was 0.74 based on 6 observations collected by 6.1-meter trawl between 2001 and 2001.

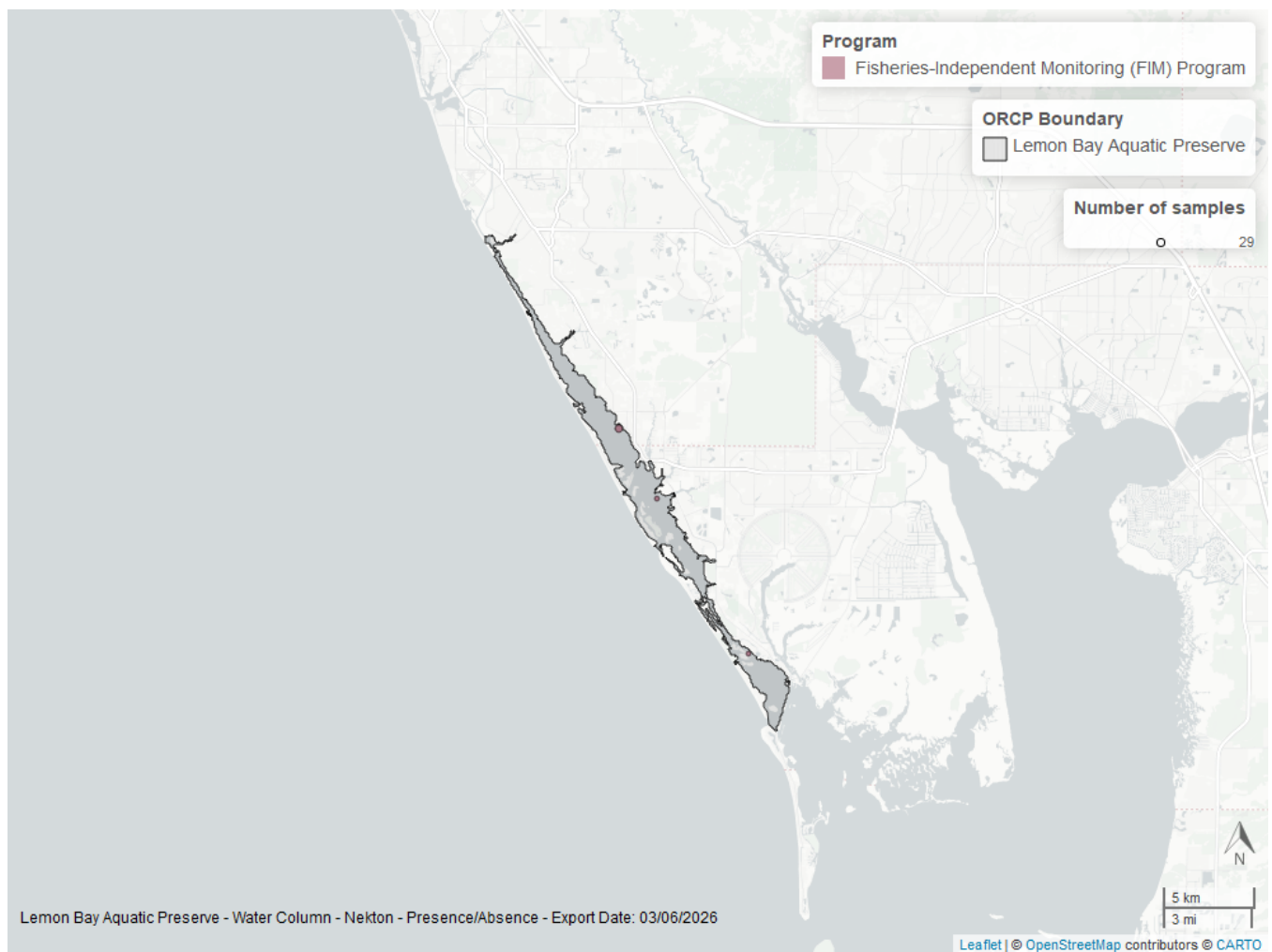


Figure 29: Map showing location of nekton sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Oyster

The data file used is: **All_OYSTER_Parameters-2026-Jun-04.txt**

Oyster sample locations available in SEACAR by Program and Year



Click [here](#) to view spatio-temporal plots on GitHub.

Density

For natural reefs, density decreased by an average of 29.56 oysters per square meter per year.

Natural

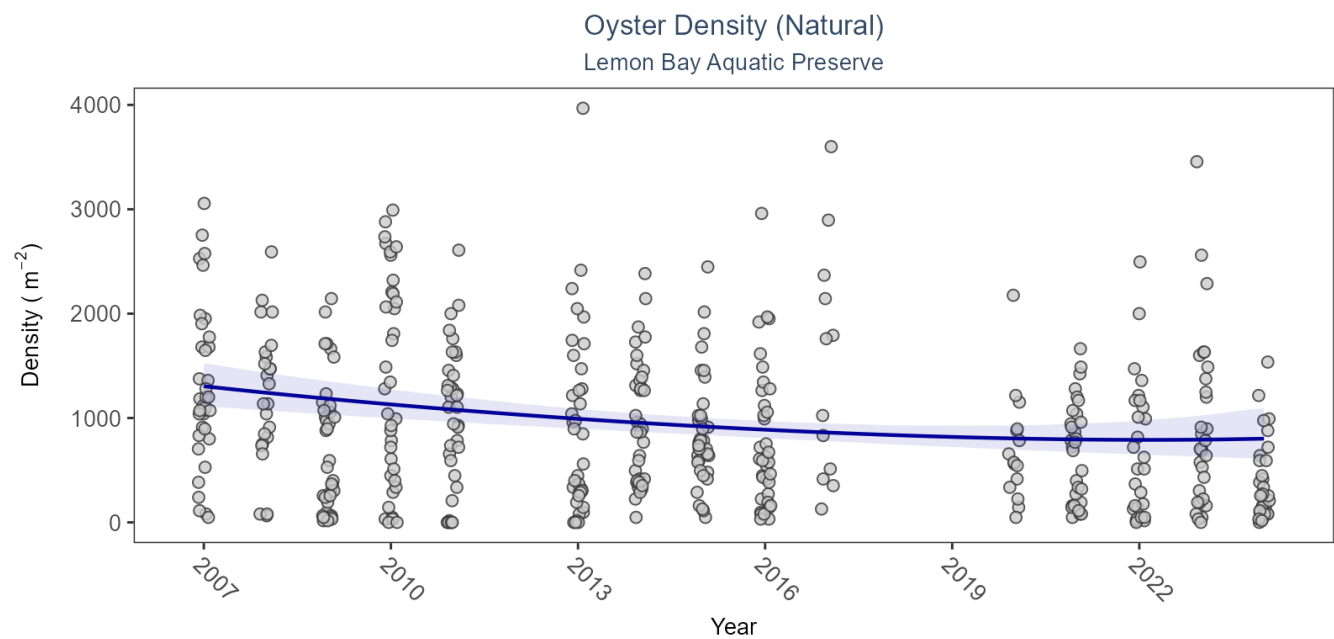


Figure 31: Scatter plot of oyster density over time. If the time series included five or more years with observations, an estimated trend (blue line) and a 95% credible interval (purple band) may also be plotted. Data points are jittered horizontally to reduce overlap.

Table 31: Model results for Oyster Density - Natural

<i>Habitat Type</i>	<i>Shell Type</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>Estimate</i>	<i>Standard Error</i>	<i>Credible Interval</i>
Natural	Live Oysters	Decreasing trend	2007 - 2024	-29.56	10.41	-49.75 to -6.83

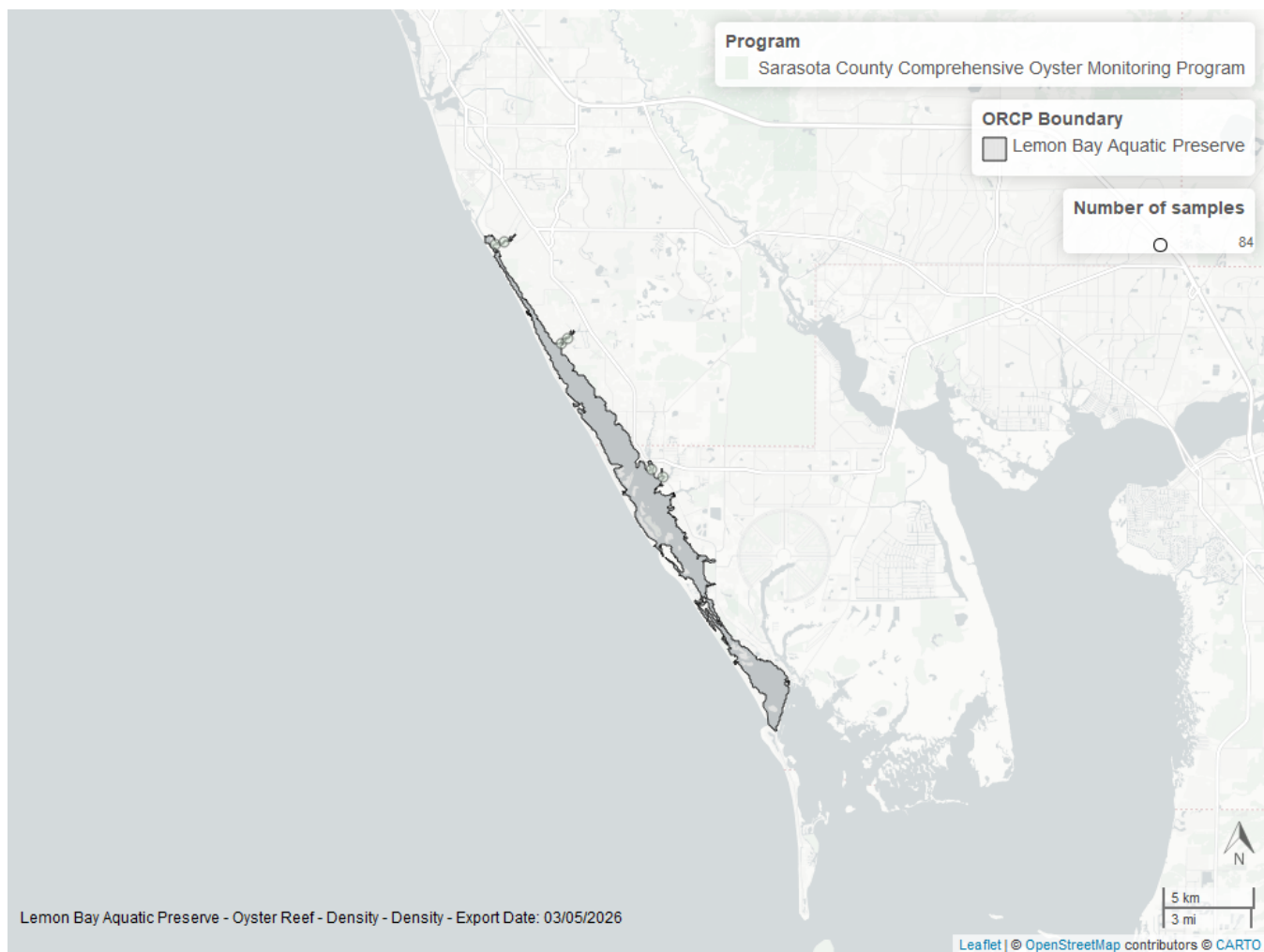


Figure 32: Map showing location of oyster density sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Percent Live

For natural reefs, percent live cover showed no detectable trend between 2007 and 2024.

Natural

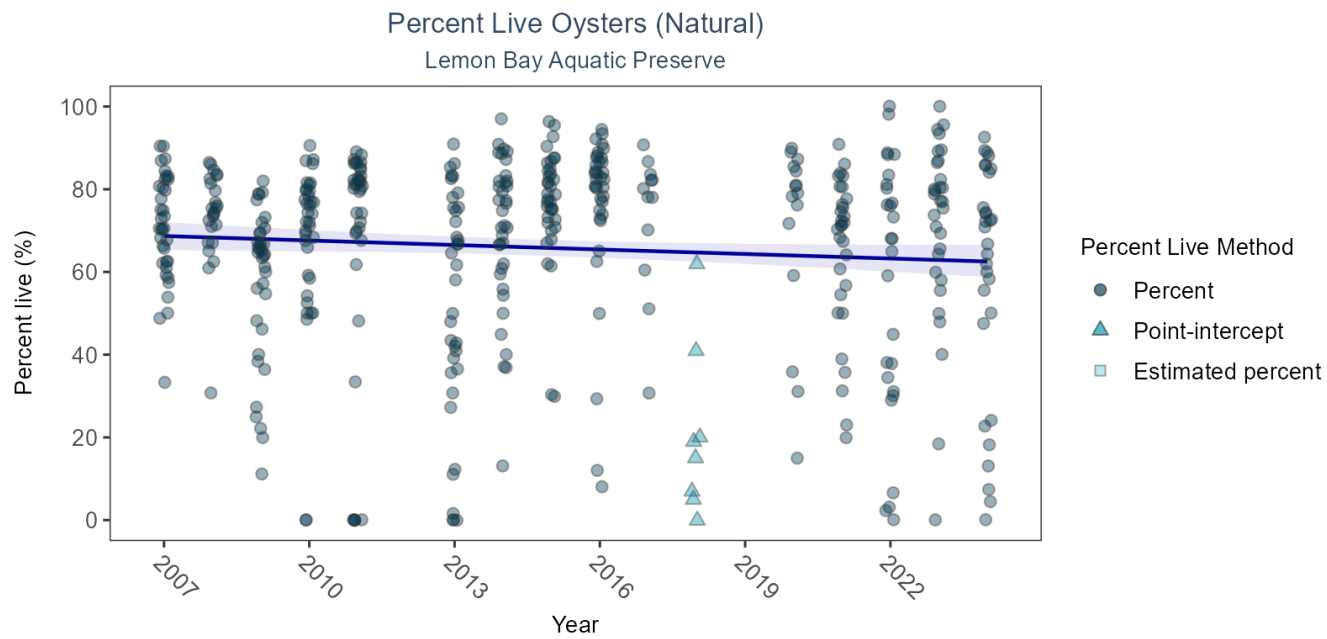


Figure 33: Scatter plot of percent live oysters over time. If the time series included five or more years with observations, an estimated trend (blue line) and a 95% credible interval (purple band) may also be plotted. Estimation method is represented as percent (circles), point-intercept (triangle), or estimated percent (square), with data points jittered horizontally to reduce overlap.

Table 32: Model results for Oyster Percent Live - Natural

<i>Habitat Type</i>	<i>Shell Type</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>Estimate</i>	<i>Standard Error</i>	<i>Credible Interval</i>
Natural	Live Oysters	No detectable trend	2007 - 2024	-0.36	0.19	-0.71 to 0.01

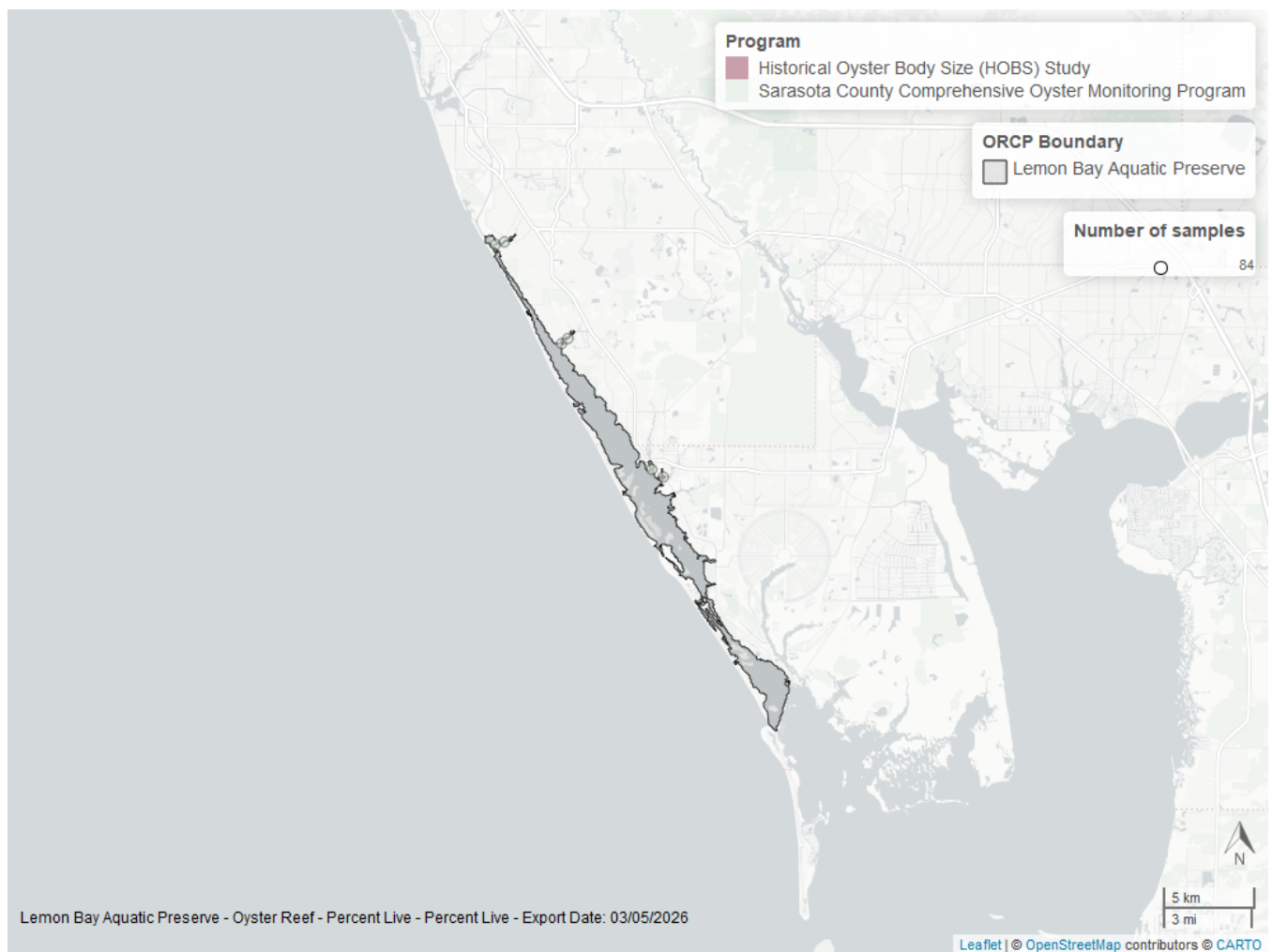


Figure 34: Map showing location of oyster percent live sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Shell Height

For natural reefs, there was insufficient data to calculate a trend for live oysters in the 25-75mm size class. Models are not run on dead oyster shell measurements.

Natural

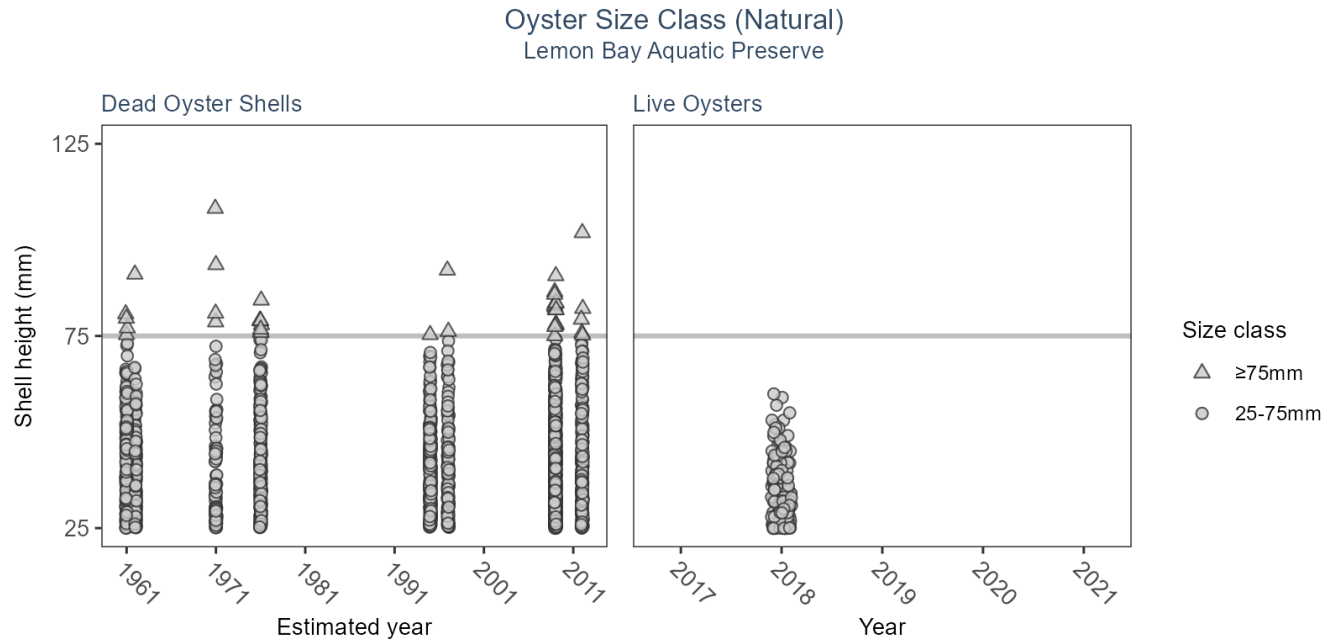


Table 33: Model results for Oyster Shell Height - Natural

<i>Habitat Type</i>	<i>Shell Type</i>	<i>SizeClass</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>No. of Samples</i>	<i>Estimate</i>	<i>Standard Error</i>	<i>Credible Interval</i>
Natural	Dead Oyster Shells	>75mm	Model not run on dead oyster shell	1961 - 2012	38	-	-	-
Natural	Dead Oyster Shells	25-75mm	Model not run on dead oyster shell	1961 - 2012	2029	-	-	-
Natural	Live Oysters	25-75mm	Insufficient data	2018 - 2018	152	-	-	-

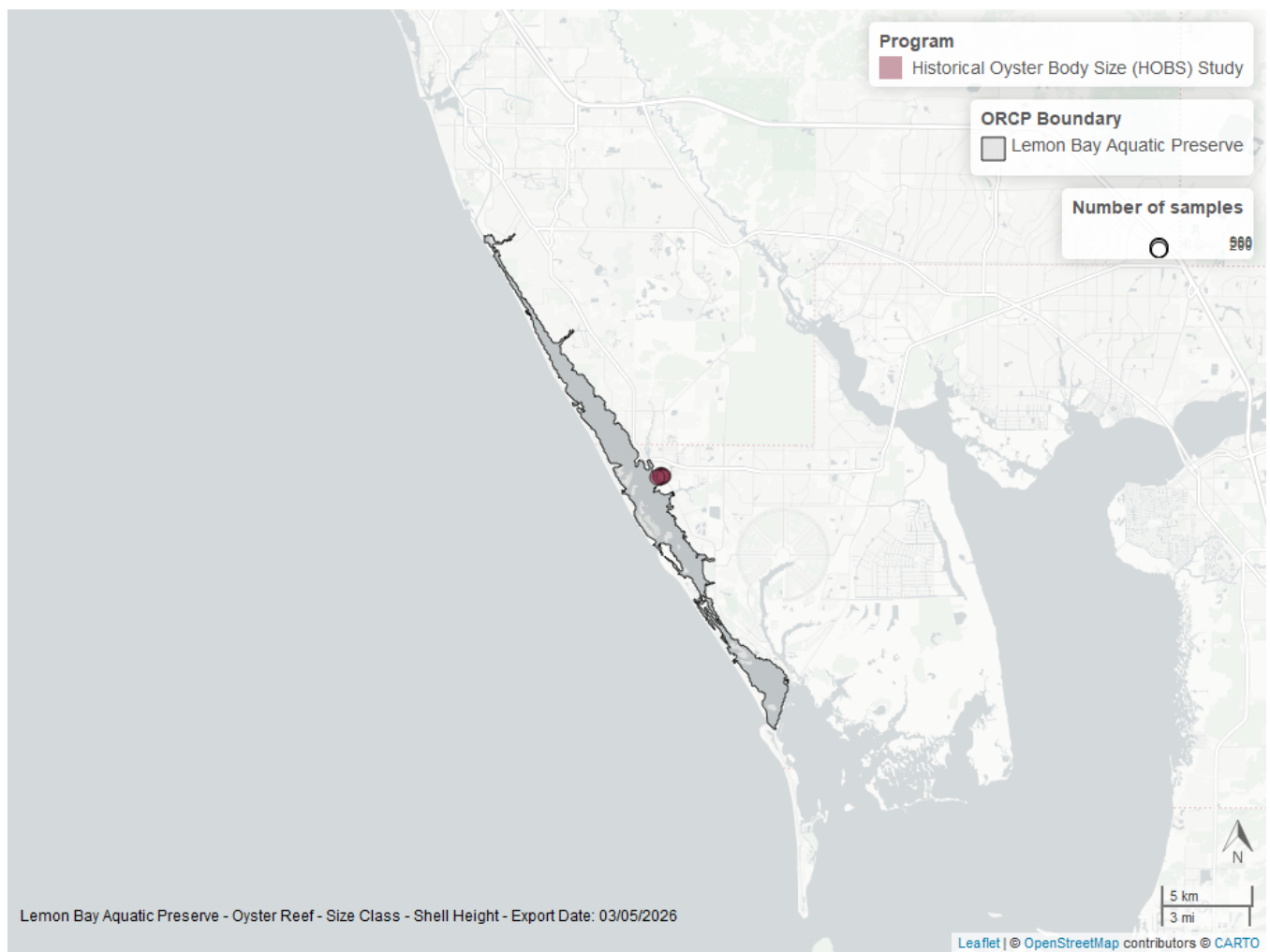


Figure 35: Map showing location of oyster shell height sampling locations within the boundaries of *Lemon Bay Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Species list

Acanthostracion quadricornis ²	Eucinostomus gula ²	Ocyurus chrysurus ²
Achirus lineatus ²	Eucinostomus harengulus ²	Ogcocephalus cubifrons ²
Albula spp. ²	Eucinostomus spp. ²	Ogcocephalus spp. ²
Aluterus schoepfii ²	Eugerres plumieri ²	Oligoplites saurus ²
Anarchopterus criniger ²	Fistularia tabacaria ²	Opisthonema oglinum ²
Anchoa hepsetus ²	Fundulus similis ²	Opsanus beta ²
Anchoa mitchilli ²	Gobiesox strumosus ²	Orthopristis chrysoptera ²
Anchoa spp. ²	Gobiosoma longipala ²	Paralichthys albigutta ²
Ancylosetta quadrocellata ²	Gobiosoma robustum ²	Penaeus duorarum ²
Archosargus probatocephalus ²	Gobiosoma spp. ²	Penicillus capitatus ¹
Ariopsis felis ²	Haemulon plumieri ²	Pogonias cromis ²
Bagre marinus ²	Halodule wrightii ¹	Prionotus rubio ²
Bairdiella chrysoura ²	Halophila engelmannii ¹	Prionotus scitulus ²
Bathygobius soporator ²	Harengula jaguana ²	Prionotus tribulus ²
Brevoortia spp. ²	Hemiramphus brasiliensis ²	Sciaenops ocellatus ²
Calamus penna ²	Hippocampus erectus ²	Scomberomorus maculatus ²
Callinectes sapidus ²	Hippocampus zosterae ²	Scorpaena brasiliensis ²
Caranx hippos ²	Hypoleurochilus caudovittatus ²	Selene vomer ²
Caranx latus ²	Hyporhamphus meeki ²	Serraniculus pumilio ²
Caulerpa mexicana ¹	Hyporhamphus unifasciatus ²	Serranus subligarius ²
Caulerpa prolifera ¹	Hypsoblennius hentz ²	Sphoeroides nephelus ²
Caulerpa sertularioides ¹	Lachnolaimus maximus ²	Sphoeroides spengleri ²
Caulerpa spp. ¹	Lagodon rhomboides ²	Sphyraena barracuda ²
Centropomus undecimalis ²	Leiostomus xanthurus ²	Strongylura marina ²
Centropristis striata ²	Limulus polyphemus	Strongylura notata ²
Chaetodipterus faber ²	Lucania parva ²	Symphurus plagiusa ²
Chasmodes saburrae ²	Lutjanus analis ²	Syngnathus floridae ²
Chilomycterus schoepfii ²	Lutjanus griseus ²	Syngnathus louisianae ²
Chloroscombrus chrysurus ²	Lutjanus synagris ²	Syngnathus scovelli ²
Citharichthys macrops ²	Menippe spp. ²	Syngnathus springeri ²
Cynoscion arenarius ²	Menticirrhus americanus ²	Synodus foetens ²
Cynoscion nebulosus ²	Menticirrhus saxatilis ²	Syringodium filiforme ¹
Diapterus auratus ²	Microgobius gulosus ²	Thalassia testudinum ¹
Diodon holocanthus ²	Monacanthus ciliatus ²	Total seagrass ¹
Diplectrum formosum ²	Mugil cephalus ²	Trachinotus carolinus ²
Diplodus holbrookii ²	Mugil curema ²	Trachinotus falcatus ²
Drift algae ¹	Mugil trichodon ²	Trinectes maculatus ²
Elops saurus ²	Mycteroperca bonaci ²	Tylosurus crocodilus ²
Epinephelus itajara ²	Mycteroperca microlepis ²	Urophycis floridana ²
Epinephelus morio ²	Nicholsina usta ²	Acanthostracion quadricornis ²
Etropus crossotus ²	No grass in quadrat ¹	Achirus lineatus ²

1 - Submerged Aquatic Vegetation, 2 - Nekton

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